SE(3)-Equivariant Dual Quaternion Graph Neural Networks for Human Motion Prediction

Chrysis Andreou

MSc Thesis Defense | University of Cyprus | June 2026

Supervisor: **Andreas Aristidou**

Table of Contents

1. Introduction
2. Background
3. Related Work
4. Method
5. Experimental Setup
6. Results
7. Discussion
8. Conclusion

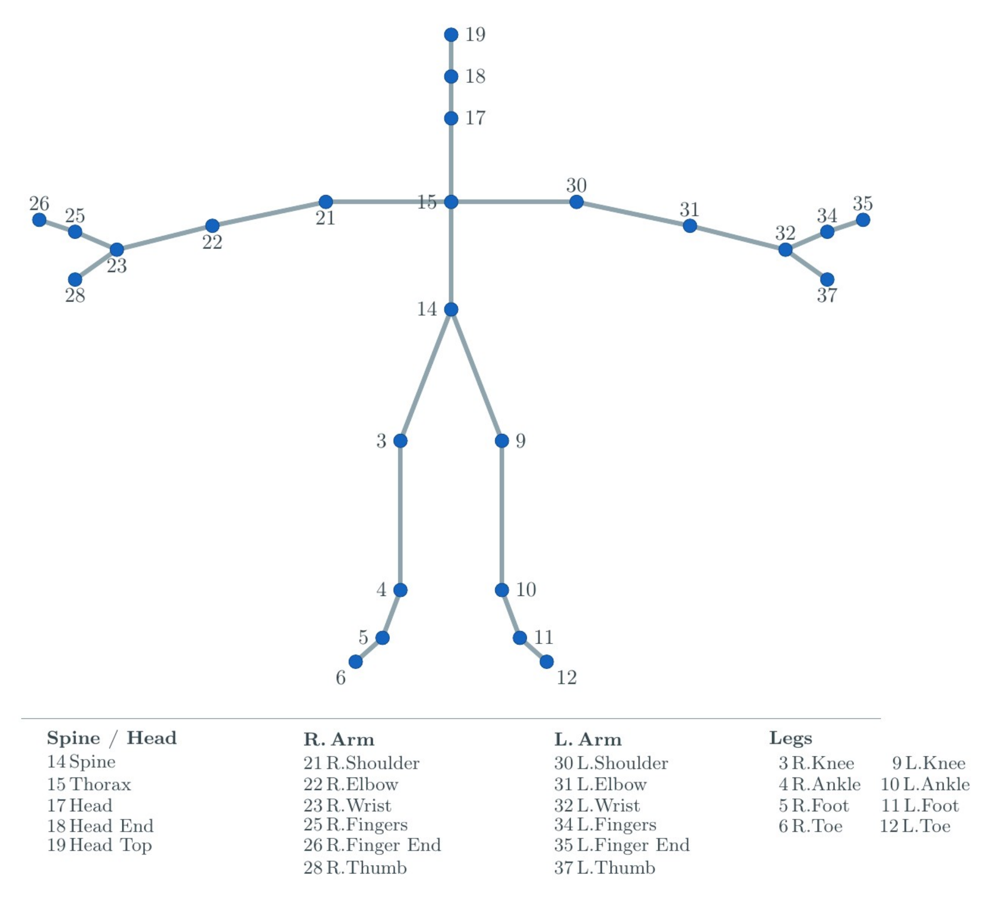
1 Introduction

1.1 Human Motion Prediction Is a Skeleton Forecast

Human motion prediction forecasts future body poses from a short observation of past motion.

A **skeleton** is a graph.

Joints are nodes. **Bones** are edges.



1.2 The Hidden Issue Is the Coordinate Frame

Standard methods represent each joint as a **3D position** in the world frame.

Rotating or translating the world frame changes the input numbers.

(For example, moving the capture sensor.)

The **physical motion is unchanged**: the person is still walking the same way.

The model may still produce a different prediction.

Standard benchmark evaluation does not expose this.

1.3 Our Approach: SE(3) Equivariance by Construction

Goal: a graph layer whose output transforms exactly with the input frame.

Two ingredients are needed.

Value type. A unit dual quaternion carries the full SE(3) symmetry (rotation plus translation).

Layer template. Equivariant graph message passing whose updates preserve the symmetry.

We combine both with the **GGMotion** backbone to build **GGMotion-DQPose.**

1.4 Contributions and Scope

1. **Reproduction of the GGMotion backbone.**

GGMotion-XYZ as a controlled position-based baseline. **35.3 mm** at 400 ms, within **0.6 mm** of the published 34.7 mm.

2. **A DQ-native GGMotion counterpart.**

GGMotion-DQPose canonicalises by torso pose, maps relative poses to se(3) log coordinates, predicts future relative coordinates, decodes through DQ reconstruction. **41.8 mm** at 400 ms.

3. **Robustness under SE(3) frame change.**

Test-time rotation up to **180°** and translation up to **5 m** without retraining. DQPose stays at 41.8 mm. XYZ rises to **47.8 mm** (rotation) and **2074 mm** (5 m translation).

Scope. CMU 8-action benchmark at 25 fps, 400 ms horizon, LTD / PGBIG protocol.

2 Background

2.1 What Background We Need

Dual quaternions give a value type for SE(3).

Graph neural networks give the message-passing framework.

Equivariance is the symmetry property we want.

SE(3) is the group of rigid motions we restrict to.

2.2 The Two Halves of a Rigid Motion

A rigid motion is a **rotation** followed by a **translation**.

Standard practice stores them **separately**: rotation matrix or unit quaternion, plus translation vector.

Composition is awkward. Rotate the second translation by the first rotation, add to the first translation, multiply the rotations.

Long kinematic chains require constant resynchronisation between rotation and translation parts.

We want **one object, one multiplication, full rigid motion**.

2.3 Quaternions Encode 3D Rotation

A **quaternion** is $q = w + xi + yj + zk$ with rules $i^2 = j^2 = k^2 = ijk = -1$.

A **unit quaternion** parameterises a 3D rotation:

$q_r = (\cos(\theta/2),\ n \sin(\theta/2))$ with axis n and angle θ.

It acts on a 3D vector by **sandwich product**: $p' = q_r\ \tilde{p}\ q_r^*$.

Unit quaternions **double-cover SO(3)**: q_r and $-q_r$ encode the same rotation.

2.4 Dual Numbers Are Numbers with a Tag

A **dual number** is $r + d\,\varepsilon$ with r real, d dual, and rules $\boldsymbol{\varepsilon^2 = 0}$, $\varepsilon \neq 0$.

Multiplication uses the rule to drop second-order terms:

$$(r_1 + d_1\,\varepsilon)(r_2 + d_2\,\varepsilon) = r_1 r_2 + (r_1 d_2 + d_1 r_2)\,\varepsilon$$

The structure parallels complex multiplication, with $\boldsymbol{\varepsilon^2 = 0}$ instead of $i^2 = -1$.

This nilpotent property lets us **carry rotation and translation in one object** without the two parts interfering.

2.5 Dual Quaternions Pack a Full Rigid Pose

$$\hat{q} = q_r + \varepsilon\, q_d$$

$\hat{q}$ a dual quaternion (8 components).

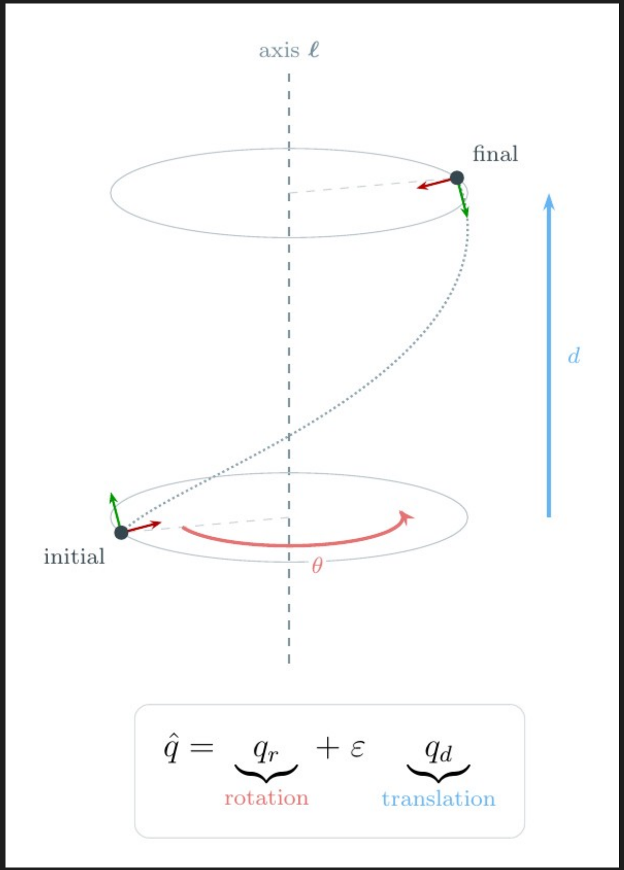
q_r real part: unit quaternion for rotation.

q_d dual part: encodes translation coupled to rotation.

ε the dual unit with $\varepsilon^2 = 0$.

Intuition. One algebraic object. One pose token.

Why this matters. Every joint pose becomes one number knowing orientation and position. Composing two motions is one multiplication.



2.6 The Translation Encoding

$$q_d = \tfrac{1}{2}\, \tilde{t}\, q_r, \qquad \text{with } \tilde{t} = (0, t)$$

Variables.

t the 3D translation vector to encode.

$\tilde{t}$ t embedded as a pure-imaginary quaternion (scalar part zero).

q_r the rotation part of the same pose.

Intuition. Translation is not stored independently. It is multiplied with the rotation through the algebra. This coupling is what allows one multiplication to compose rigid motions correctly.

Why this matters. This rule turns a translation vector into the right object to live alongside a rotation in one DQ.

2.7 One Multiplication Composes a Full Rigid Motion

$$\hat{q}_1 \otimes \hat{q}_2 = q\_r1\, q\_r2 + (\, q\_r1\, q\_d2 + q\_d1\, q\_r2 \,)\, \varepsilon$$

Variables.

Real part **q_r1 q_r2** is the quaternion product of the two rotations.

Dual part **q_r1 q_d2 + q_d1 q_r2** mixes each rotation with the other's translation.

Intuition. This single rule does the bookkeeping that matrix and split storage have to do by hand. The dual-part mixing is exactly the rotate-the-translation-then-add step that composes rigid motions correctly.

Why this matters. Composition equals one multiplication. The relative pose between two joints is one multiplication: $\tilde{q} = \hat{q}\_c^{-1} \otimes \hat{q}\_i$. This is the operation that the equivariance proof uses.

2.8 Log and Exp Map DQs to a Flat 6D Space

$$\xi = \log(\tilde{q}) \in se(3) \cong R^6, \qquad \tilde{q} = \exp(\xi)$$

Variables.

$\tilde{q}$ a unit dual quaternion on a curved manifold.

$\xi = (\omega, v)$ a twist: 6D vector with 3 rotation generators (ω = axis × angle) and 3 translation generators (v).

Intuition. Unit DQs are constrained. The log map flattens a relative pose into ordinary R^6 where MLPs and attention work normally. The exp map maps the network's R^6 output back to a valid pose.

Why this matters. The neural network operates on flat 6D twists while the geometric meaning lives in DQs. Log gives the network a learnable coordinate; exp gives it a way to produce valid poses.

2.9 Graph Neural Networks Propagate Information

A GNN processes objects (joints) related by pairwise connections (bones).

Each layer has three steps.

1. **Edge message**. $m_{ij} = \varphi_e(h_i, h_j, a_{ij})$. Learned function of the two incident joints.
2. **Aggregation**. $m_i = \mathrm{Agg}_j\, m_{ij}$. Permutation-invariant operator (sum or mean).
3. **Node update**. $h_i^{(\ell+1)} = \varphi_h(h_i, m_i)$. Next-layer state.

Permutation-equivariant by construction.

No geometric equivariance to rotations or translations. That is what we add next.

2.10 The Equivariance Principle

$$f(\rho_X(g)\, x) = \rho_Y(g)\, f(x)$$

Variables.

f the model.

g a group element (a frame change).

$\rho_X(g)$ how g acts on the input.

$\rho_Y(g)$ how g acts on the output.

Intuition. A function is equivariant when applying a transformation to its input is the same as applying that transformation to its output. For motion prediction: rotating the input pose should rotate the predicted future pose.

Why this matters. The symmetry we want by construction. Three payoffs: sample efficiency, out-of-distribution robustness, parameter capacity used efficiently.

2.11 EGNN: The Equivariant Coordinate Update

$$x_i^{(\ell+1)} = x_i^{(\ell)} + (1 / |N(i)|) \Sigma_j (x_i - x_j) \varphi_x(m_{ij})$$

Variables.

x_i equivariant position of node i.

$(x_i - x_j)$ neighbour-relative difference.

$\varphi_x(m_{ij})$ scalar weight computed from invariant edge features.

Intuition. Each position is updated by a sum of neighbour-relative directions, each scaled by a learned scalar. Differences make translations cancel; scalars commute with orthogonal transformations. Equivariance is built into the update form.

Why this matters. Template for every equivariant graph layer. GGMotion-DQPose adopts this template, replacing 3D positions with 6D twists.

2.12 GMN Handles Rigid Constraints

EGNN models free particles. Each position updates independently.

Skeletons have **rigid constraints**: bone lengths cannot change.

GMN works in **generalised coordinates** for each rigid substructure.

Cartesian states are recovered through a **forward-kinematics map** that satisfies the constraints by construction.

GGMotion sits in this lineage.

GGMotion-DQPose inherits the constraint-aware structure and replaces the position axis with the DQ log axis.

2.13 The Group We Care About: SE(3)

$$SE(3) = \{\ (R, t)\ :\ R \in SO(3),\ t \in \mathbb{R}^3\ \}$$

Variables.

$R \in SO(3)$ a 3D rotation matrix.

$t \in \mathbb{R}^3$ a 3D translation vector.

Action on a point: $x \mapsto R\,x + t$.

Intuition. SE(3) is the group of all rigid motions in 3D. Rotation followed by translation. No scaling, no shearing, no reflection. Every frame change is an element of SE(3).

Why this matters. Skeletal motion is generated by joint rotations and limb translations. Reflections are not motions a real body can perform.

2.14 SE(3) Acts on Pose DQs by Left Multiplication

Unit DQs form a **double cover** of SE(3). $\hat{q}$ and $-\hat{q}$ encode the same rigid motion.

The SE(3) action on a pose is **left multiplication**: $g \cdot \hat{q} = g \otimes \hat{q}$.

The Lie algebra **se(3)** (tangent space at identity) is 6-dimensional.

The **exp / log map** gives a smooth chart between se(3) and a neighbourhood of identity in SE(3).

This is what lets the network output 6D vectors that map back to valid poses.

2.15 Human Motion Prediction Activates All Four Ingredients

Joints are **vertices**. Bones are **edges**. Graph topology is fixed across frames.

A rigid motion of the world frame should produce the same predicted trajectory in the rotated and translated frame.

Evaluation metric: **MPJPE in millimetres**. Mean Euclidean distance between predicted and ground-truth joint positions.

All ingredients align with the CMU protocol: graph topology, equivariance target, evaluation surface.

3 Related Work

3.1 Three Threads, Three Gaps

Motion prediction methods. Sequence models on per-joint trajectories. Graph-based spatial reasoning (LTD, MSR-GCN, PGBIG). Attention variants (SANet, DA-MgTCN, StaAttention).

Gap. Cartesian positions throughout. No SE(3) guarantee.

Equivariant GNNs for motion. EGNN to GMN to GGMotion lineage. Sister branches: ESTAG, EGNO, EqMotion, SEGNO, PAINET.

Gap. Position-only representation. Orientation never represented as a first-class quantity.

Dual quaternions in deep learning. DQ-MLP and DQ-LSTM-VAE (Vieira et al.). Quaternion GNNs (QPU, RE-QGCN, QSTGNN, QMP).

Gap. Either no graph, or quaternion-only (rotation but not translation).

3.2 The Lineage We Build On

EGNN > **GMN** > **GGMotion** > **GGMotion-DQPose**

EGNN (Satorras et al., 2021): canonical E(n)-equivariant message-passing layer.

GMN (Huang et al., 2022): generalises EGNN to constraint-aware systems through generalised coordinates and forward kinematics.

GGMotion (Wan et al., 2025): extends GMN with hierarchical group and part attention. Reports 34.7 mm at 400 ms on CMU. Strongest published equivariant method.

GGMotion-DQPose (this thesis): replaces the $\mathbb{R}^3$ position axis with the SE(3) twist axis.

3.3 Positioning of This Thesis

The literature has **two of three ingredients** at a time, never all three.

DQ thread. Algebra. No graph.

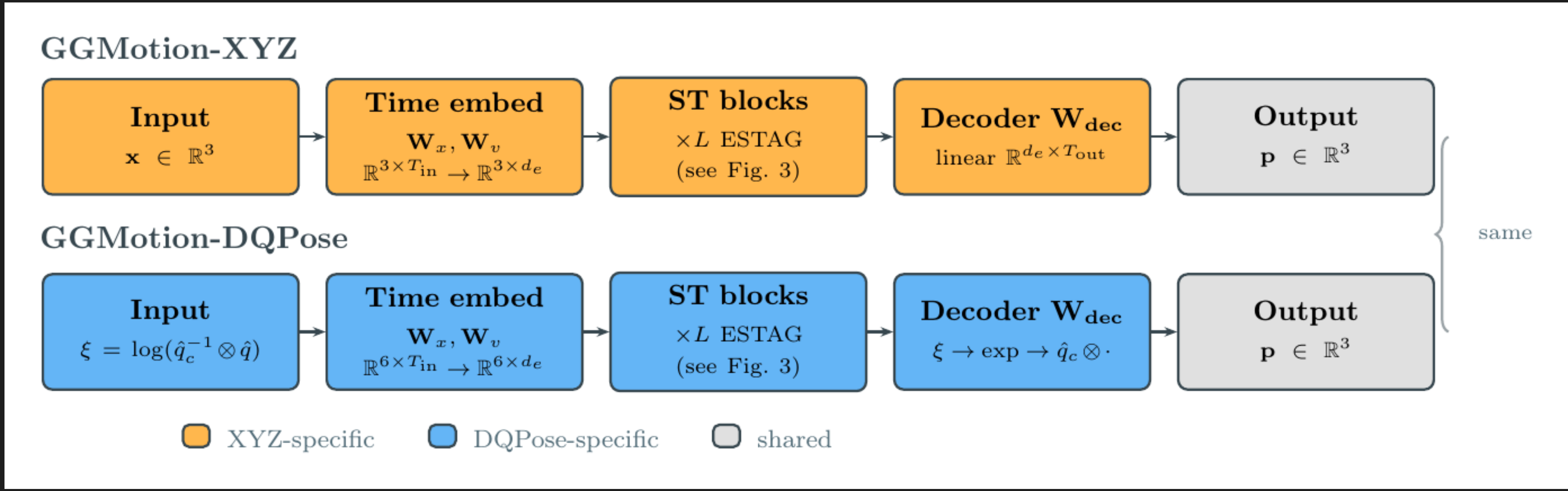
Quaternion-GNN thread. Graph. Only rotation.

Equivariant-GNN thread. Graph and symmetry. Position-only.

This thesis closes all three gaps simultaneously. A DQ-native graph backbone with SE(3) equivariance by construction, evaluated on the standard CMU benchmark.

4 Method

4.1a Pipeline: GGMotion-XYZ



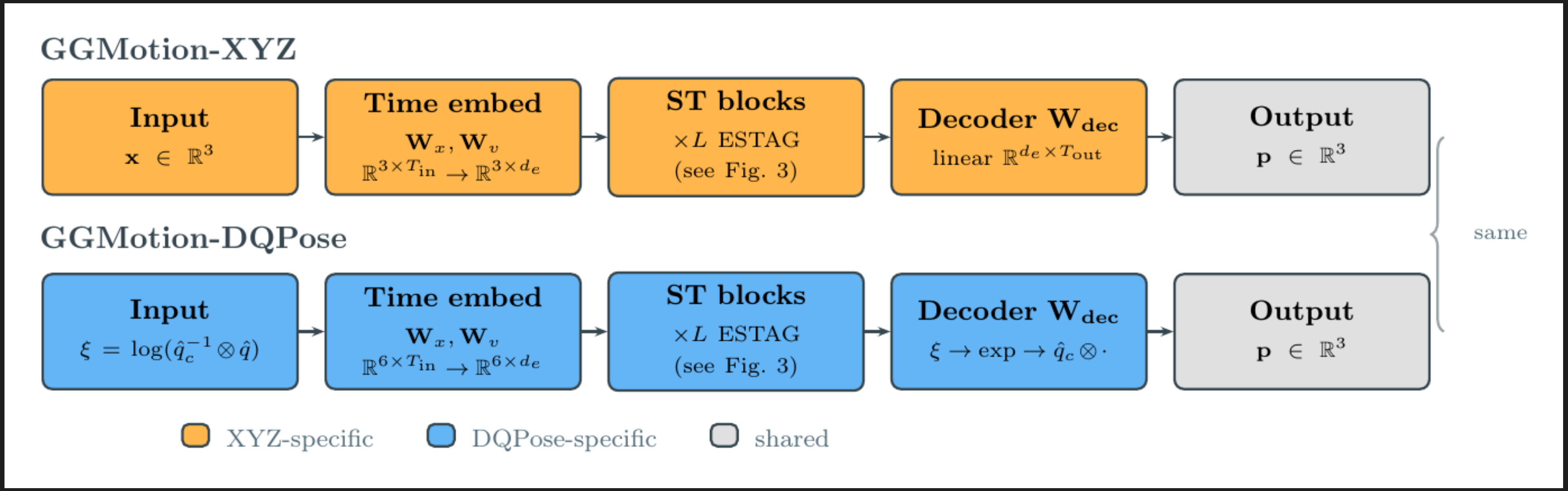
Input. Raw CMU motion converted to global joint positions by forward kinematics.

Internal state. 3D Cartesian positions $x_i \in R^3$ per joint, per frame.

Backbone. Four spatio-temporal GGMotion blocks (spatial / temporal / group / part attention). Per-component 3D scales (x, y, z).
spatial = neighbors on the graph, temporal = motion over time, group = between limbs, part = inside each limb.

Decoder. Linear projection from feature axis to predicted future positions. MPJPE directly.

4.1b Pipeline: GGMotion-DQPose



Input. Raw CMU motion. FK produces global rotation and global position per joint. Each joint becomes a pose DQ.

Preprocessing. Use the last torso pose as reference. Each joint becomes a pose relative to the torso ($\tilde{q}\_i = \hat{q}\_c^{-1} \otimes \hat{q}\_i$), then a 6D log vector ($\xi\_i = \log(\tilde{q}\_i) \in R^6$) for the backbone. The network never sees absolute world coordinates — only how each joint moves with respect to the torso.

Backbone. Same four GGMotion blocks. 6D log coordinate stream. Scalar per-joint scales.

Decoder. Predict relative motion in 6D → exp to a pose → multiply by the stored torso reference $\hat{q}\_c$ → read joint positions → MPJPE.

4.2 Forward Kinematics Gives Each Joint a Full Pose

$$\hat{q}\_i\^{}(t) = r\_i\^{}(t) + \varepsilon \cdot \tfrac{1}{2}\, t\_i\^{}(t) \;\square\; r\_i\^{}(t)$$

Variables.

$R\_i\^{}(t)$ $\in$ **SO(3)**, **$p\_i\^{}(t)$** $\in$ **R^3**: global rotation and global position of joint i at frame t, from FK.

$r\_i\^{}(t)$: unit quaternion for the global rotation.

$t\_i\^{}(t) = (0, p\_i\^{}(t))$: position embedded as a pure-imaginary quaternion.

Intuition. Raw CMU data stores relative rotations to parents plus a root translation. We chain these root-to-leaf to recover each joint's global pose. Each global pose is packed into one pose DQ.

Why this matters. XYZ uses only the position. DQPose uses the full pose (rotation plus position). First step that gives the network access to orientation as a first-class quantity.

4.3 The Reference Pose Removes the World Frame

$$\tilde{q}_i^{(t)} = \hat{q}_c^{-1} \otimes \hat{q}_i^{(t)}, \qquad \xi_i^{(t)} = \log(\tilde{q}_i^{(t)}) \in \mathbb{R}^6$$

Variables.

$\hat{q}_c = \hat{q}_8^{(T_{in})}$: last observed pose of joint 8 (the torso).

$\tilde{q}_i^{(t)}$: pose of joint i at frame t, relative to $\hat{q}_c$.

$\xi_i^{(t)}$: 6D log coordinate of that relative pose.

Intuition. The network never sees where the skeleton is in the world. It sees only how each joint relates to the torso reference. The global frame is stored in $\hat{q}_c$ (not discarded) and reintroduced algebraically at the decoder.

Why this matters. This is the canonicalisation step that makes equivariance possible.

4.4 Frame Cancellation Is the Load-Bearing Mechanism

$$(g \otimes \hat{q}_c)^{-1} \otimes (g \otimes \hat{q}_i^{(t)}) \;=\; \hat{q}_c^{-1} \otimes \hat{q}_i^{(t)}$$

Variables.

$g \in SE(3)$: any global rigid frame change (rotation, translation, or both).

$g\,\hat{q}_c$: reference pose after the frame change.

$g \otimes \hat{q}_i$: joint pose after the frame change.

Intuition. When g is applied to every pose, it appears on both sides of the relative-pose computation and cancels algebraically. The network sees an identical input regardless of the global frame.

Why this matters. This is why DQPose is exactly equivariant. The mechanism is one line of algebra, not a learned approximation. Same cancellation applies to DQ velocities. Every tensor the backbone sees is frame-independent.

4.5 Why GGMotion-XYZ Is Not SE(3)-Equivariant

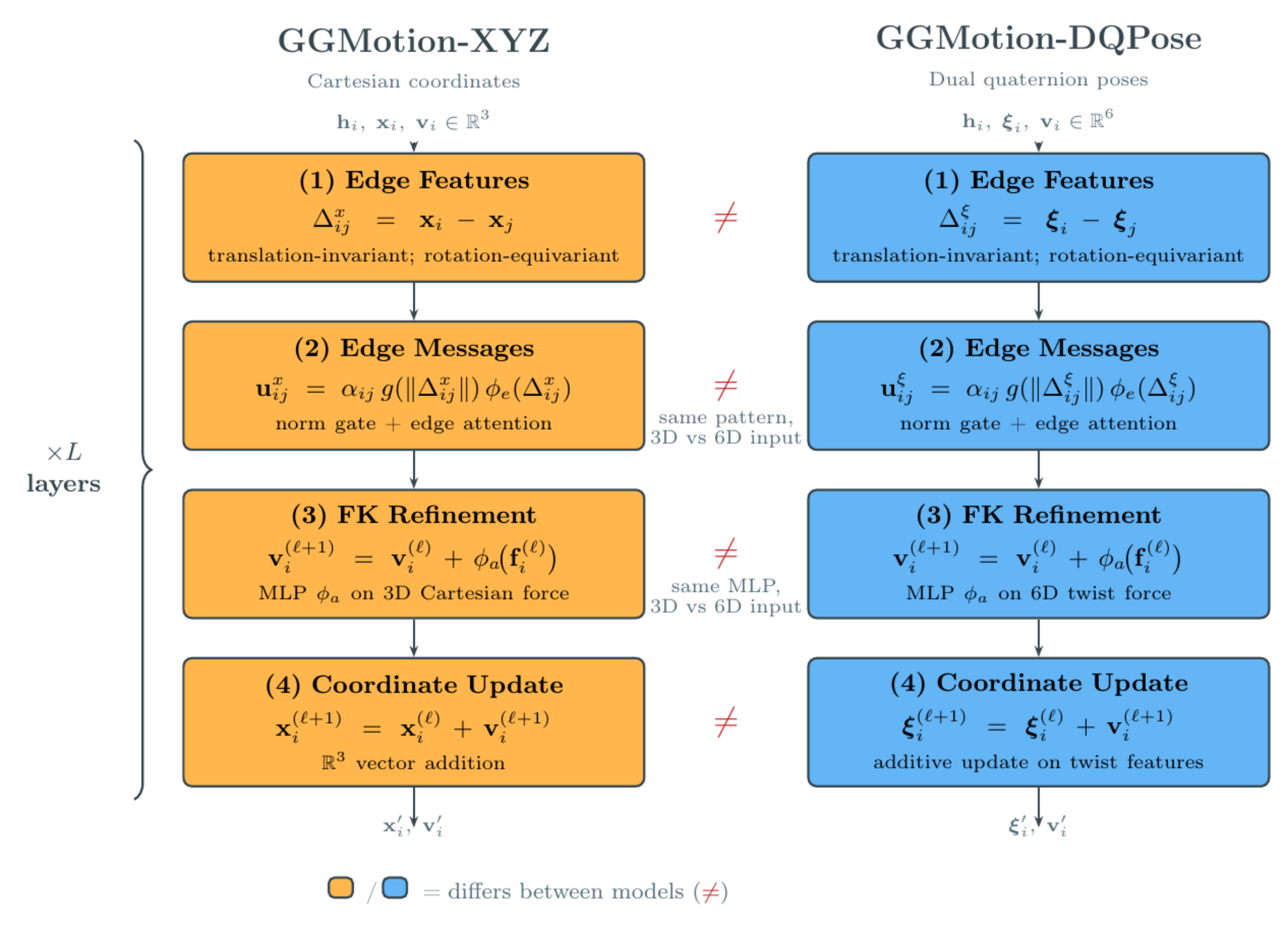
Per-component 3D scales. XYZ uses learned scales on each axis independently. Under rotation, axes mix. Per-axis scales do not commute with channel mixing by a rotation matrix.

Learned centroid projection. Centroid subtraction is a preprocessing convention, not an SE(3) action. Learned projections of feature tensors can shift differently under translation than a true SE(3) action.

Temporal residual. Absolute-frame quantities can leak through the temporal pathway.

Result. Strong canonical-frame accuracy. Function tied to the dataset frame. The robustness experiment measures this empirically.

4.6 The Backbone Operates on a 6D Log Stream



Same GGMotion block family: spatial attention, temporal attention, group / part attention.

XYZ: 3D Cartesian streams, per-component 3D scales.

DQPose: 6D twist streams, scalar per-joint scales (treats the 6D twist as a single geometric object).

Conservative design. Change the minimum required to support pose DQs. Any difference in behaviour traces to the representation.

4.7 DQ Velocity Stays in the Same 6D Language

Velocity in DQPose is not an XYZ displacement.

DQ velocity:

$$\dot{\xi}_i^{(t)} = \log\left(\hat{q}_i^{(t-1)^{-1}} \otimes \hat{q}_i^{(t)} \right) \in R^6$$

This tells the backbone how each joint pose **moves between frames** in the same 6D pose language.

The same frame cancellation applies. DQ velocity is invariant under global left-multiplication of all poses.

4.8 The Decoder Reconstructs Future Positions

$$\hat{q}_{pred}^{(t)} = \hat{q}_c \otimes \exp(\eta_i^{(t)}), \qquad \hat{p}_i^{(t)} = \mathrm{dq2pos}(\,\hat{q}_{pred}^{(t)}\,)$$

Variables.

$\eta_i^{(t)}$: network's predicted relative future 6D coordinate.

$\exp(\eta)$: a unit relative pose DQ.

$\hat{q}_c \otimes \exp(\eta)$: global predicted pose, with the reference frame reintroduced.

dq2pos: extracts the 3D translation from the dual part of the DQ.

Intuition. The backbone predicts in the flat reference-relative coordinate space. Exp turns it back into a valid pose. Composition with $\hat{q}_c$ (reference pose dual quaternion — the last observed torso pose in the input window) restores the global frame algebraically. Position is extracted only at the end, for MPJPE.

Why this matters. Encoder removes the frame. Decoder mirrors it. Algebraic restoration is what makes equivariance hold end to end.

4.9 The Training Loss Keeps the Comparison Honest

Position loss (dominant). L_2 on decoded positions. $L\_pos = (1 / NT) \Sigma \| \hat{p} - p \|_2$. Both models use it.

Bone-vector loss (auxiliary). L_1 on parent-child bone differences. Preserves skeleton structure.

DQ regularisers (DQPose only, from epoch 11). Light DQ pose geodesic and DQ velocity terms. Weights $\lambda\_dq = 0.005$, $\lambda\_vel = 0.005$.

The DQ terms are supportive. Dominant pressure is decoded-position MPJPE. Keeps both models comparable under the same metric.

4.10 SE(3) Equivariance Theorem

Theorem. Let f denote GGMotion-DQPose. Let $g \in SE(3)$ act on every input pose by left multiplication. Then $\mathbf{f(g \cdot \hat{q}) = g \cdot f(\hat{q})}$.

Proof in three steps.

1. **Cancellation.** Every relative pose passed to the network is unchanged by g (see 4.4).
2. **Backbone.** The learned tensor is identical, so the backbone predicts the same relative future twists.
3. **Decoder.** The transformed run composes predictions with the transformed reference pose, so output positions transform by g.

Empirical verification. Random SE(3) transforms applied to the input. Discrepancy at numerical precision.

5 Experimental Setup

5.1 CMU 8-Action Benchmark

Dataset. CMU Motion Capture.

Actions. basketball, basketball signal, directing traffic, jumping, running, soccer, walking, washwindow.

Protocol. 10 observed frames in, 10 predicted frames out, at 25 fps.

Horizons. 80, 160, 320, 400 ms (the 2nd, 4th, 8th, 10th predicted frames).

Split. Standard Li et al. train / test split, used unmodified.

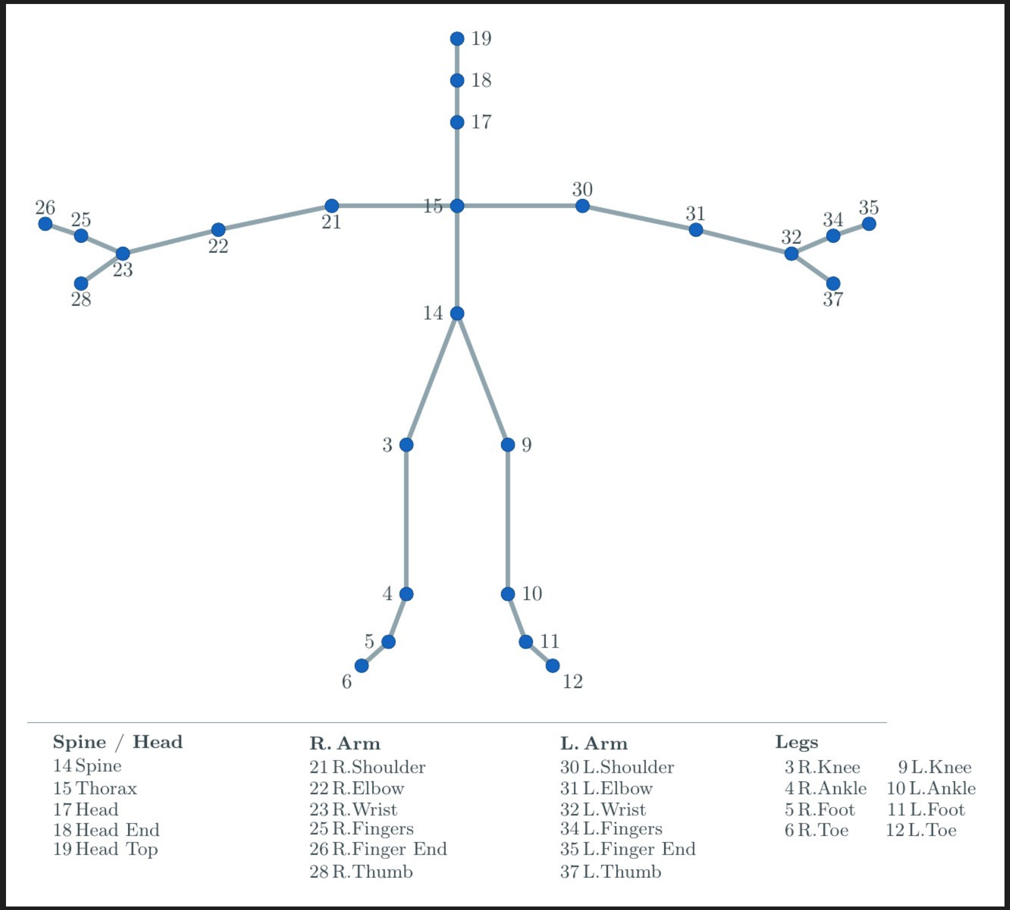
5.2 Skeleton Topology: 25 In, 38 Out

Raw CMU motion has **38 joints**.

Following LTD and PGBIG, we drop **13 auxiliary joints** (root, end effectors, intermediate spine).

Model trains and runs inference on the **25-joint subset**.

At reporting time the 13 auxiliary joints are restored (7 by duplication, 6 from ground truth) to form the **38-joint MPJPE skeleton**.



5.3 Mean Per Joint Position Error Is the Evaluation Metric

$$\text{MPJPE} = (1 / |J|) \sum_j \| \hat{x}_j - x_j \|_2$$

Variables.

J: evaluated joint set (38 joints).

$\hat{x}_j$: predicted 3D position of joint j, in millimetres.

x_j: ground-truth 3D position of joint j.

Intuition. Average distance in millimetres between where the model predicted each joint and where it actually was. Lower is better. Reported at four horizons. DQPose is evaluated by decoded positions; the metric is fair to both models.

Why this matters. Standard CMU metric. Keeps GGMotion-DQPose directly comparable to all published baselines.

5.4 Published Baselines on the Same Protocol

Method	80 ms	160 ms	320 ms	400 ms
SANet	7.1	13.2	26.8	34.1
GGMotion (published)	7.1	13.2	27.0	34.7
DA-MgTCN	7.5	14.0	28.1	34.8
PGBIG	7.6	14.3	29.0	36.6
StaAttention	6.5	14.7	29.4	37.2
MSR-GCN	8.8	15.9	30.4	37.9
GGMotion-XYZ (ours)	**7.2**	**13.3**	**27.4**	**35.3**
GGMotion-DQPose (ours)	**8.4**	**15.7**	**32.4**	**41.8**

5.5 Fairness Protocol

1. **Same backbone.** Same layer count, hidden dimension, equivariant feature dimension, graph topology, temporal embedding, group / part attention, future-window projection.

2. **Same hyperparameters.** Optimisation, data, architecture. Only representation-driven rows differ.

3. **Same benchmark objective.** Both models selected and reported by the same decoded-position MPJPE at the same four horizons.

4. **Identical evaluation.** Same script, same 25-joint output, same 38-joint MPJPE reconstruction, no model-specific adjustments.

5.6 Same Training Configuration

Optimiser. Adam. Initial LR 3×10^{-4}. Weight decay 10^{-5}.

LR schedule. Step at 19 milestones, factor 0.88 per step.

Gradient clipping. Max norm 1.0.

Seed. 1234567890 (GGMotion default).

Budget. 50 epochs.

DQPose early stop. Start 12, patience 5, $\delta = 0.05$ mm. Triggers at epoch 23.

Best checkpoints. XYZ epoch 19. DQPose epoch 22.

Hardware. NVIDIA RTX A5000 (training, RunPod). Apple M1 Max CPU (evaluation, equivariance, robustness).

6 Results

6.1 Canonical Accuracy: XYZ Is Stronger

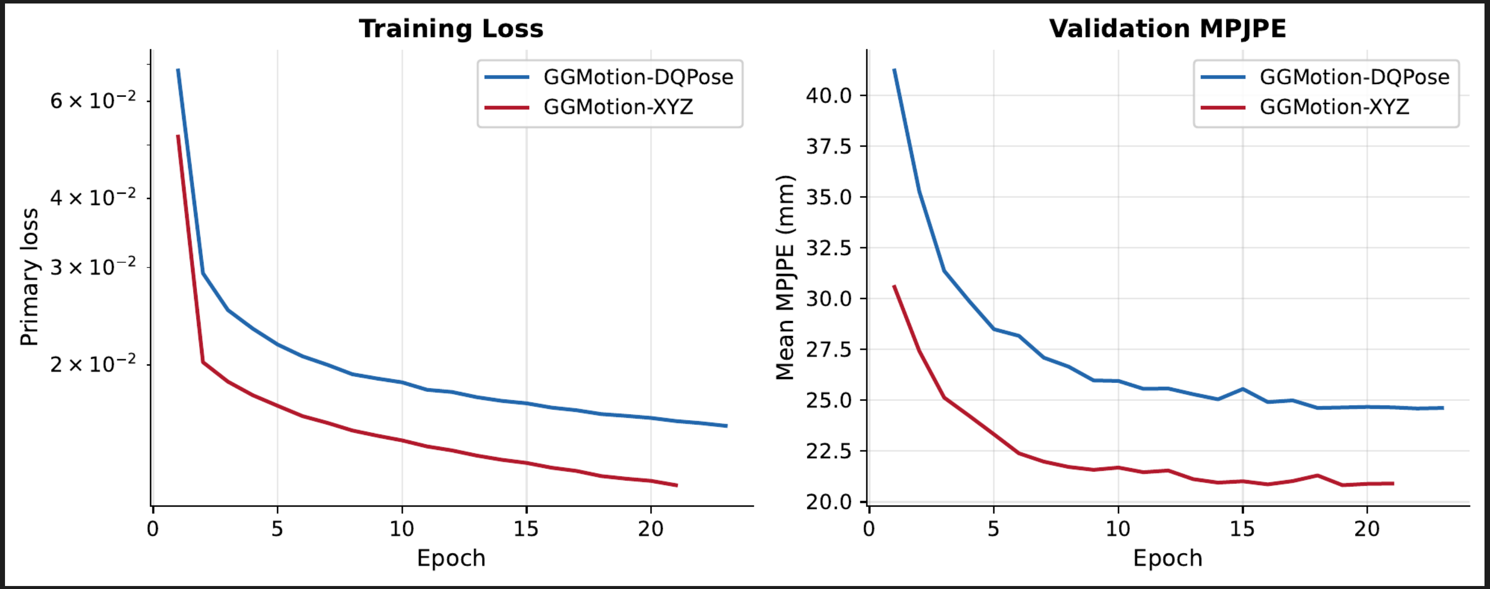
Model	80 ms	160 ms	320 ms	400 ms
GGMotion-XYZ	7.2	13.3	27.4	**35.3**
GGMotion-DQPose	8.4	15.7	32.4	**41.8**

GGMotion-XYZ reaches **35.3 mm** at 400 ms, within 0.6 mm of the published 34.7 mm.

GGMotion-DQPose reaches **41.8 mm**: a **6.5 mm gap**.

The gap grows with horizon. Canonical MPJPE favours XYZ. Robustness is next.

6.2 Training Convergence: Both Models Learn



Both models reach a **stable validation region**.

Neither model collapses.

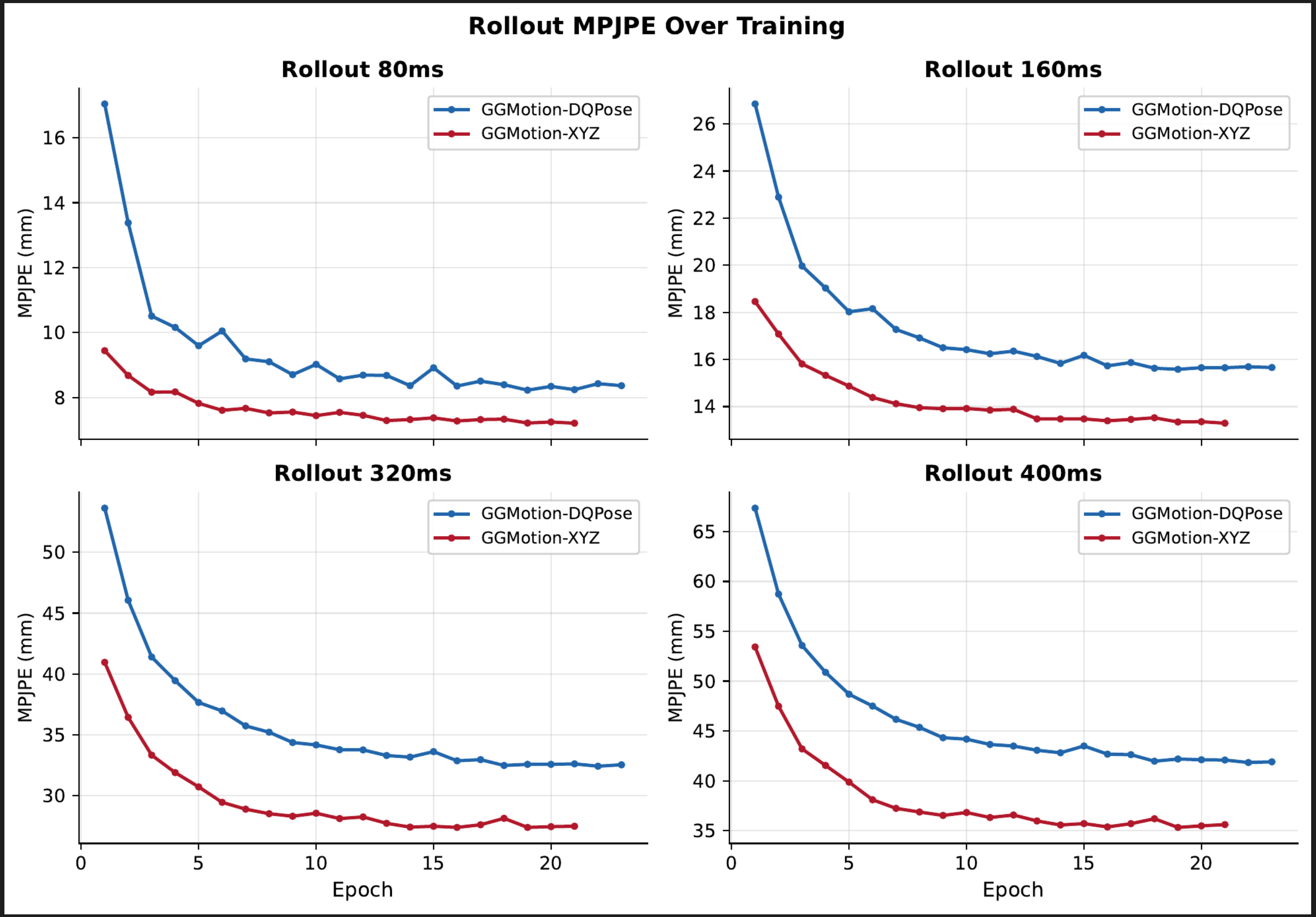
DQPose remains above XYZ throughout.

Reported checkpoints:

XYZ epoch 19

DQPose epoch 22

6.3 Rollout Convergence by Horizon



Each curve: validation MPJPE per prediction horizon.

DQPose gap is present at **every horizon** throughout training.

The gap **grows at longer horizons**, matching the test-set pattern.

6.4 Reading the Accuracy Figures

Each action accuracy figure is a grid.

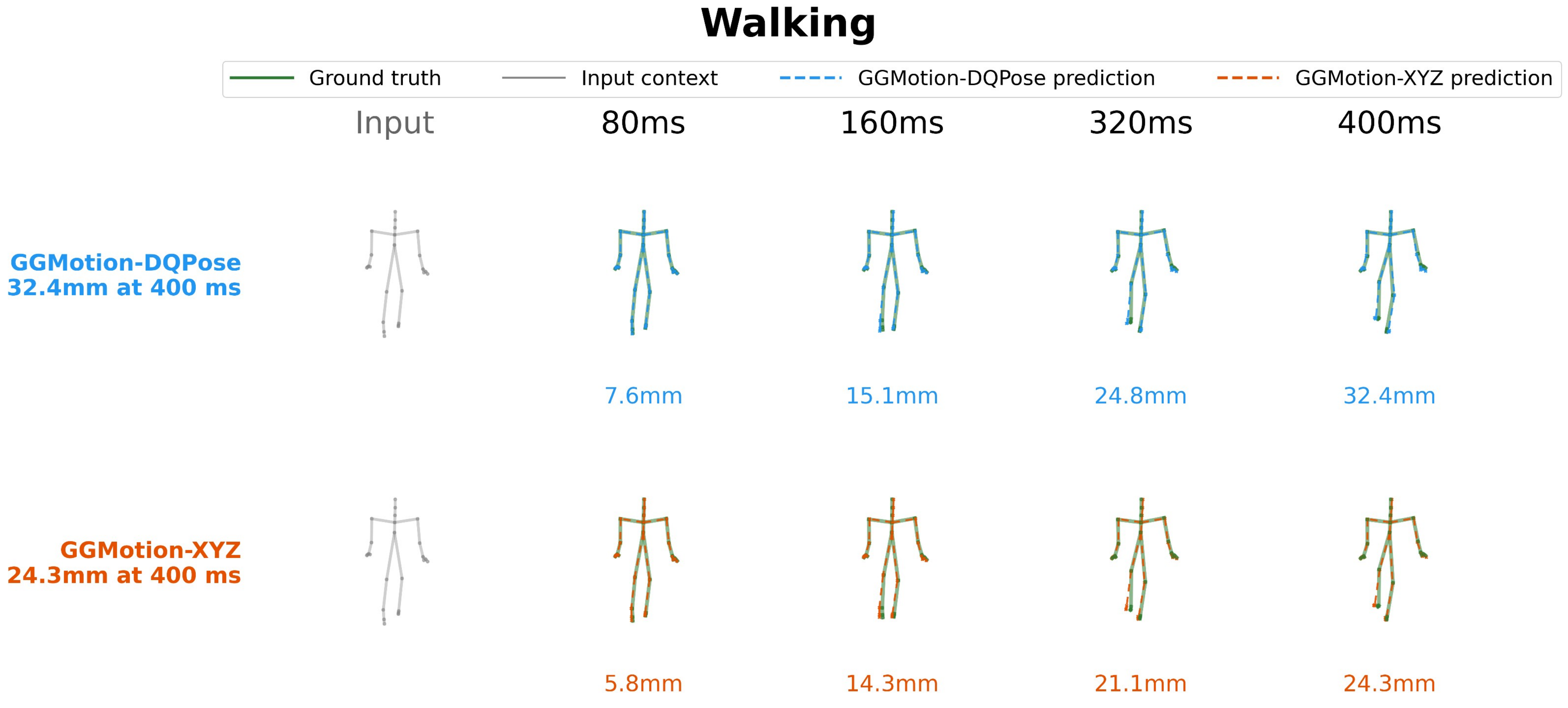
Columns: input window plus the four horizons (80, 160, 320, 400 ms).

Rows: **Model.** DQPose on top. XYZ on bottom.

Colours: green = ground truth, dashed = prediction, grey = observed input context.

Use these as qualitative evidence beside the MPJPE table.

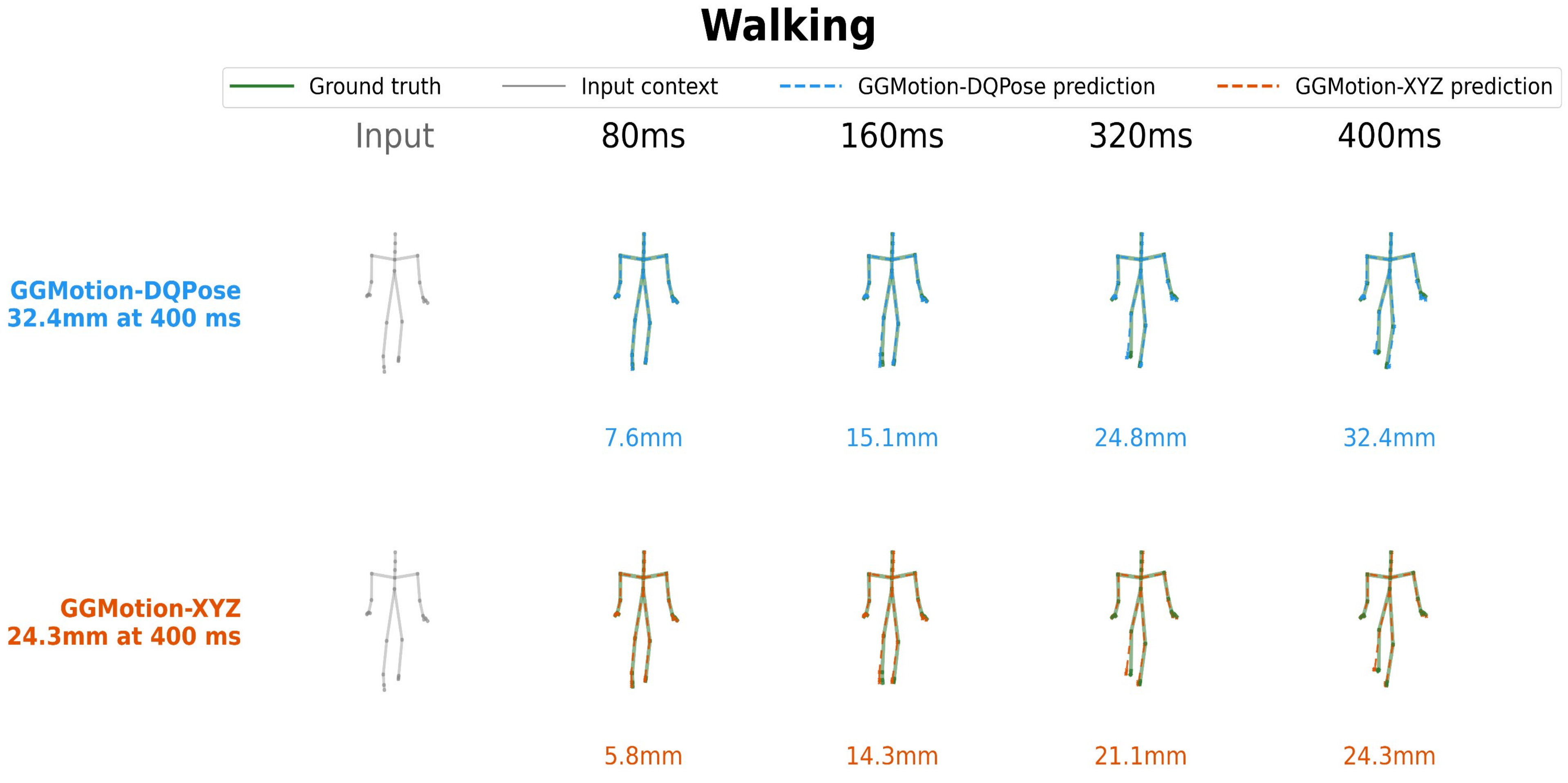
6.5 Accuracy: Walking {figure}



Walking is the **clearest representative locomotion** action.

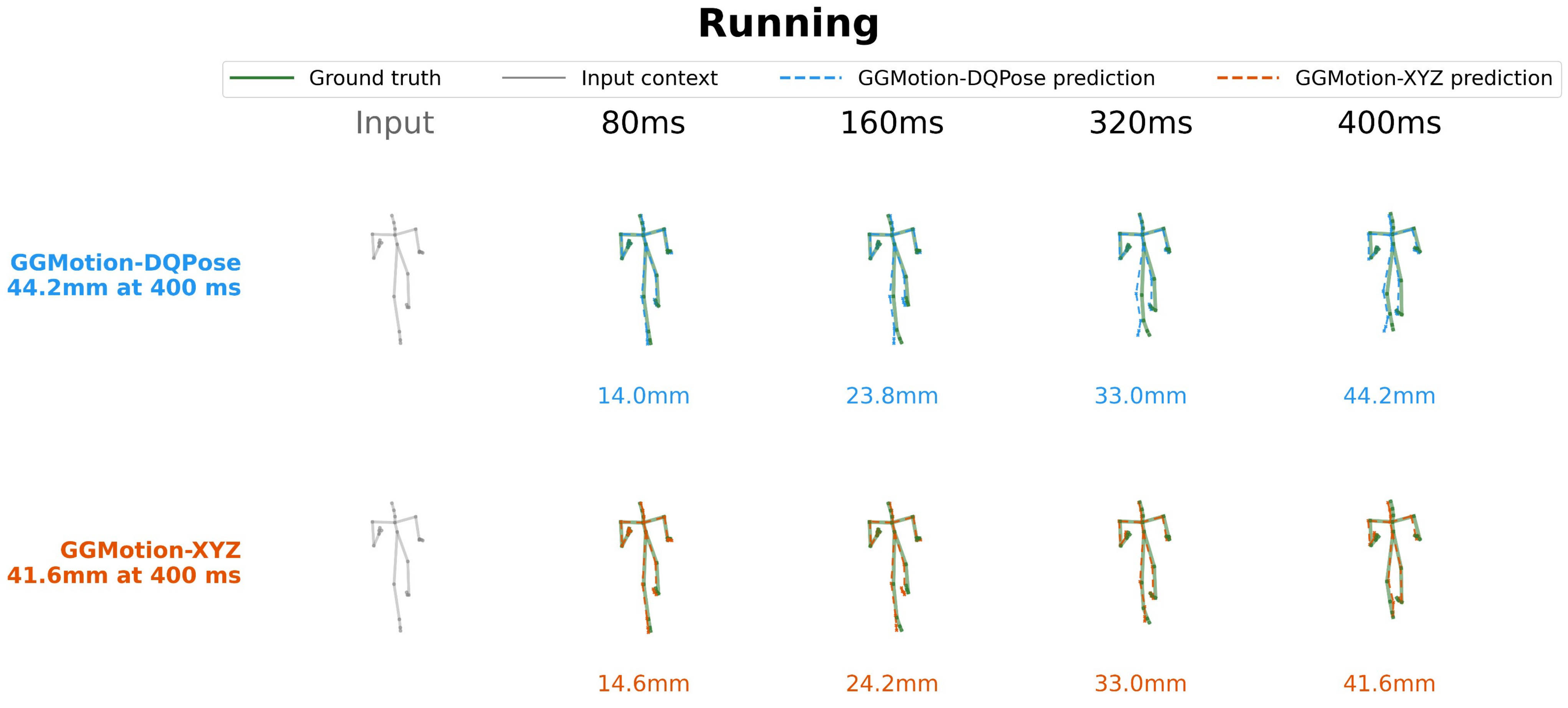
Both models keep plausible skeleton geometry. XYZ closer: 24.3 vs 32.4mm at 400 ms.

6.6 Accuracy: Walking (Video)

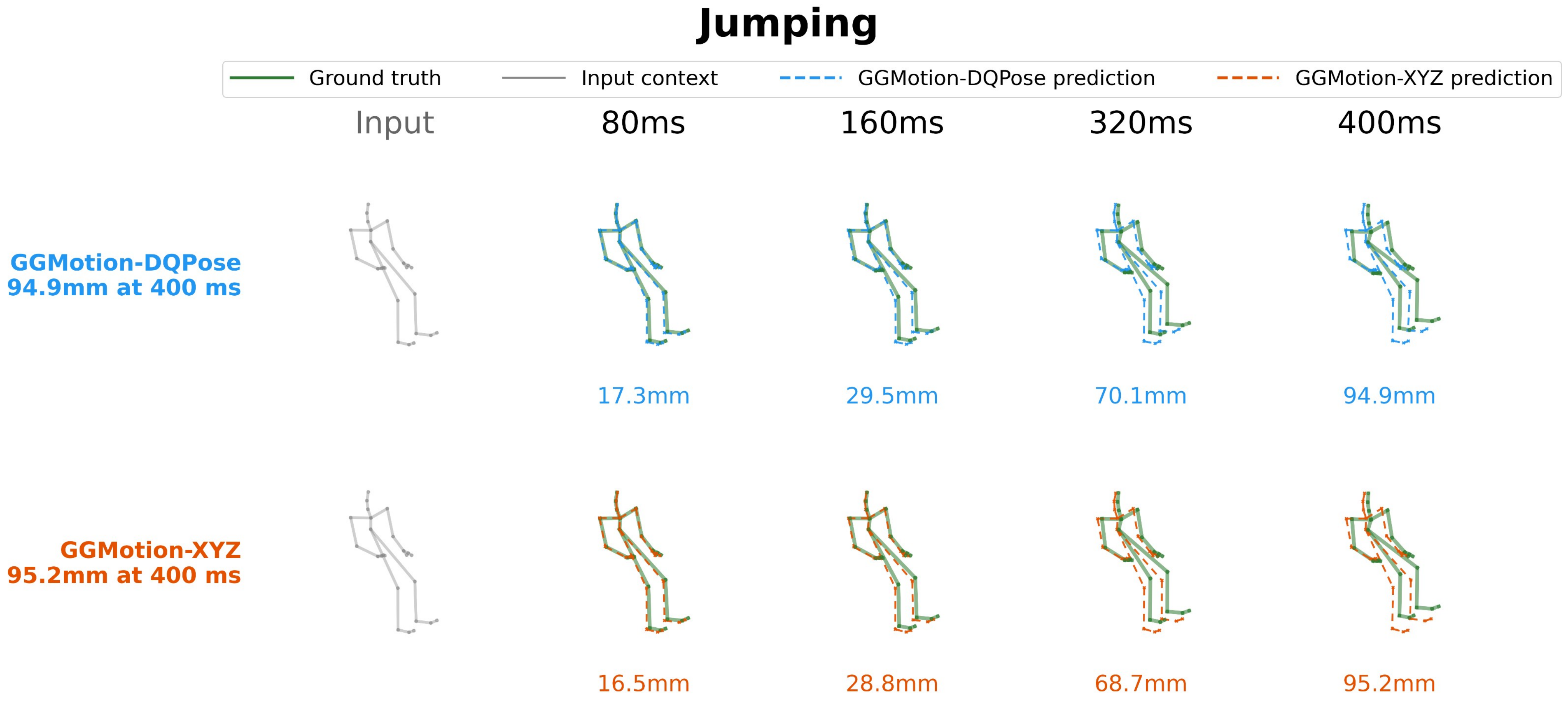


Diagnostic as motion. DQPose plays plausibly. XYZ tracks ground truth more tightly.

6.7 Accuracy: Running



6.8 Accuracy: Jumping



6.9 Perturbation Protocol for Robustness

1. Apply **SE(3) frame change** to the input. Same physical motion, different coordinate frame.

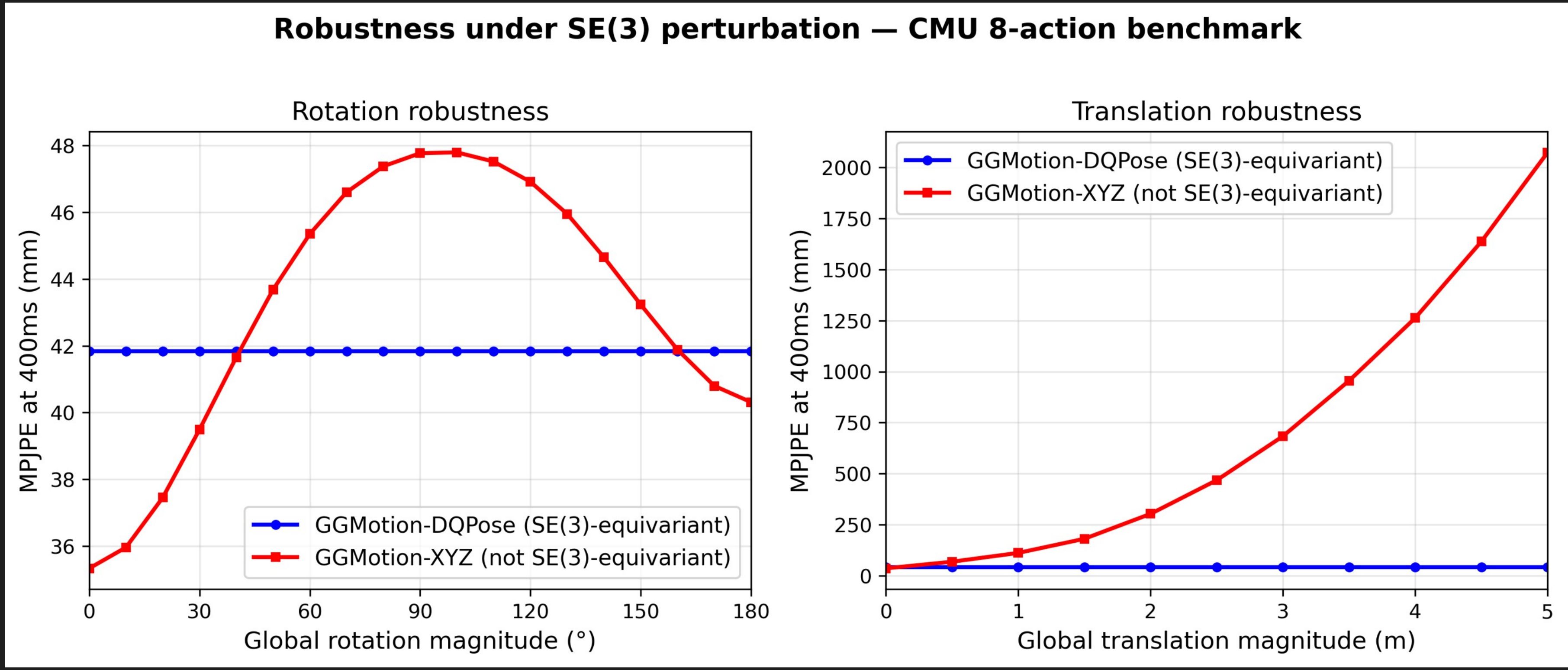
2. Predict in the changed frame. No retraining, no fine-tuning.

3. Evaluate MPJPE in the changed frame against the correspondingly transformed ground truth.

An equivariant model's MPJPE should stay **constant**.

Two sweeps. Rotation 0° to 180° in 10° steps (random axis, seed 42). Translation 0 to 5 m in 0.5 m steps (random direction, seed 99).

6.10 Main Robustness {Plot}



DQPose stays flat at 41.8 mm across the full rotation sweep and the full translation sweep.

XYZ rotation: rises to **47.8 mm** near 100°. Partial recovery to 40.3 mm at 180°.

XYZ translation: rises to **2074 mm** at 5 m. A 49.6× ratio relative to DQPose.

6.11 Translation Exposes Absolute Coordinate Dependence

A **5 m global translation** does not change the physical motion.

DQPose. Cancels the translation through the reference-relative pose path. $\hat{q}_c$ absorbs the offset.

XYZ. Exposes hidden dependence on absolute coordinate values.

Numbers:

DQPose flat at **41.8 mm**.

XYZ at 0.5 m: **67.9 mm**.

XYZ at 1 m: **111 mm**.

XYZ at 5 m: **2074 mm**.

6.12 Reading the Rotation Figures

Each rotation figure is a 3×5 grid.

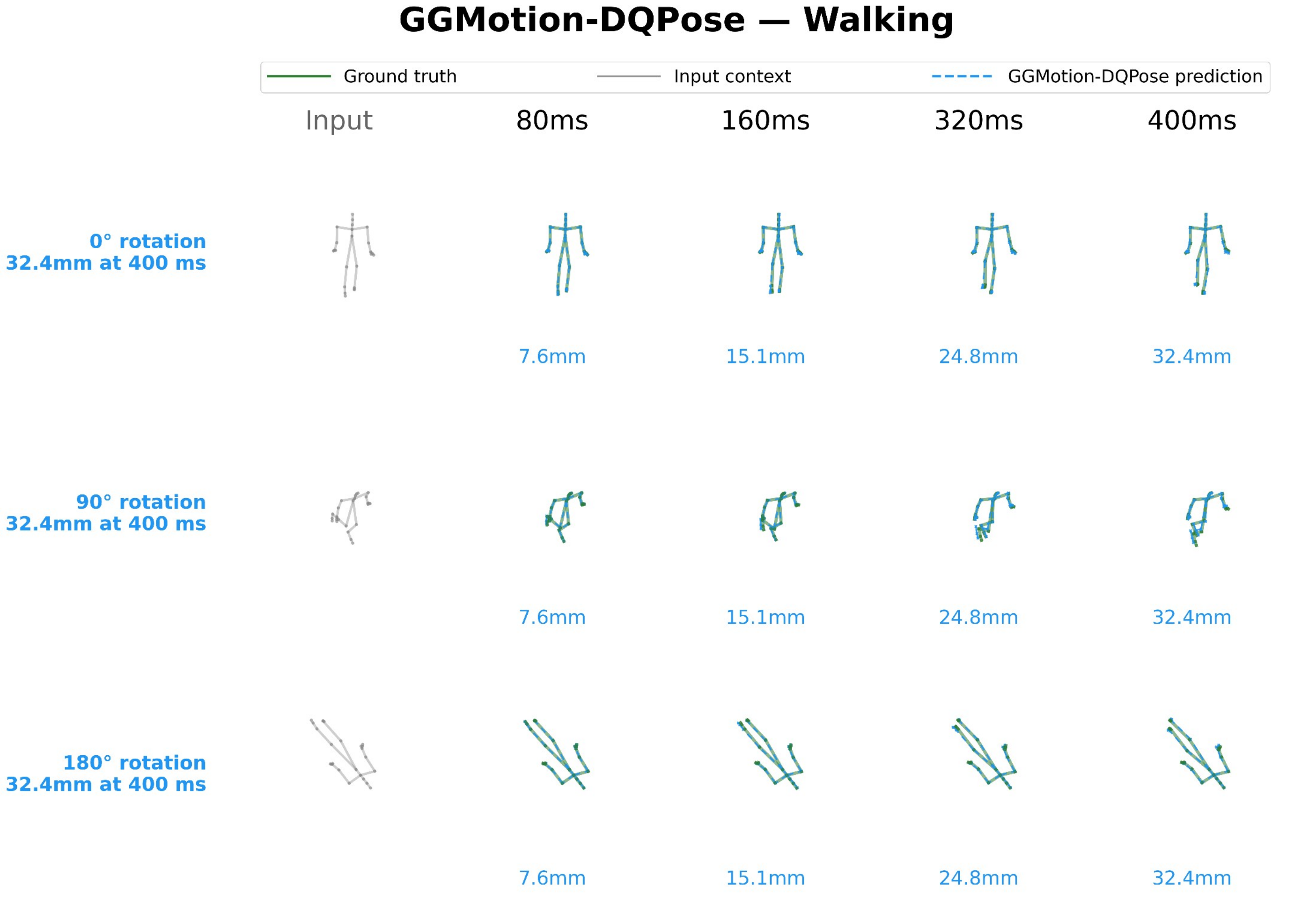
Rows: rotation angles 0°, 90°, 180°.

Columns: prediction horizons (input, 80, 160, 320, 400 ms).

DQPose rows should look like **rotated copies** of the 0° prediction.

XYZ rows may show different prediction geometry. The output changes when the same motion is expressed in a rotated frame.

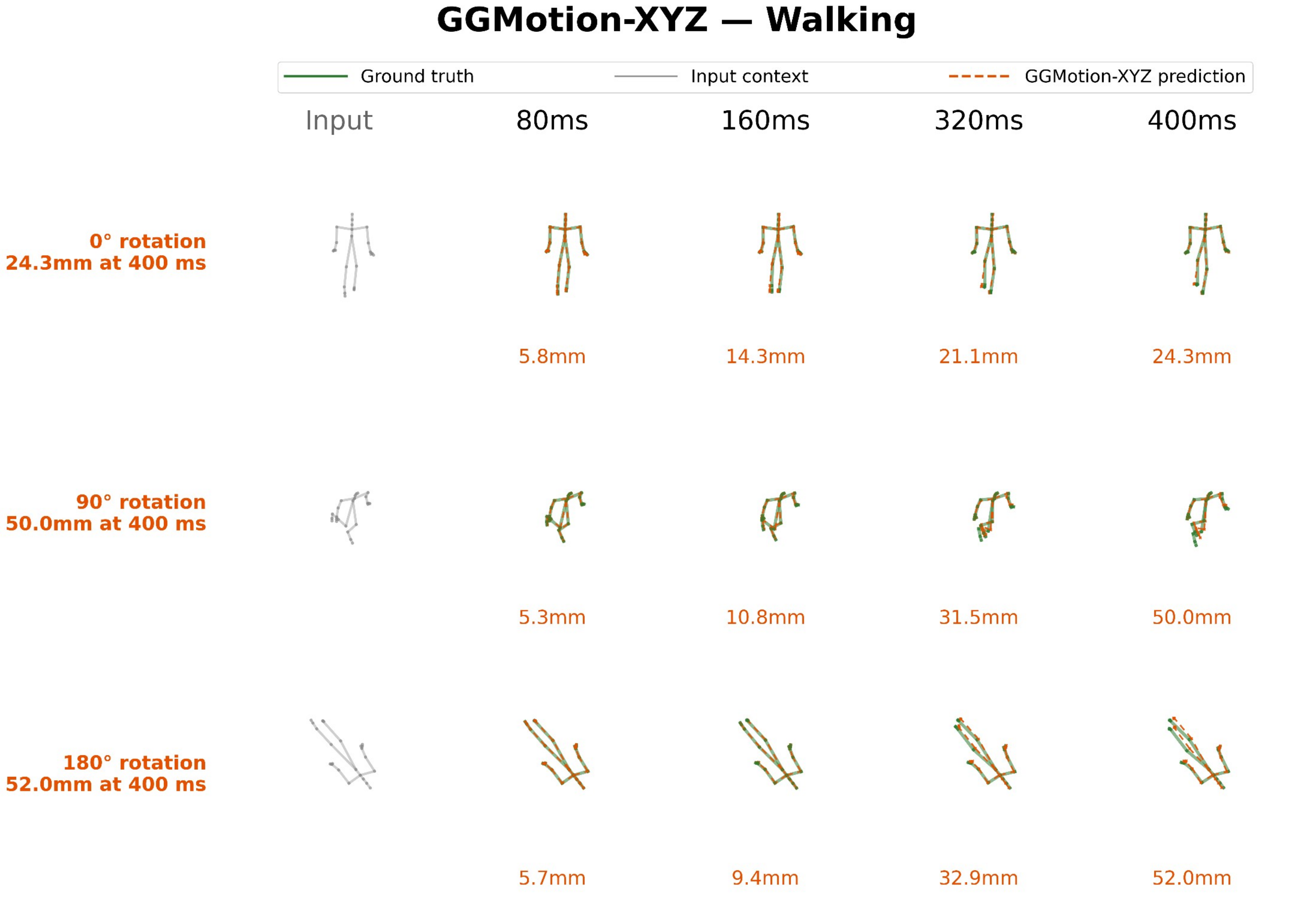
6.13a Rotation Walking: DQPose



Three rows: 0°, 90°, 180° rotation.

Predictions are **rotated copies** of the same skeleton. DQPose MPJPE constant at **32.4 mm** across all angles.

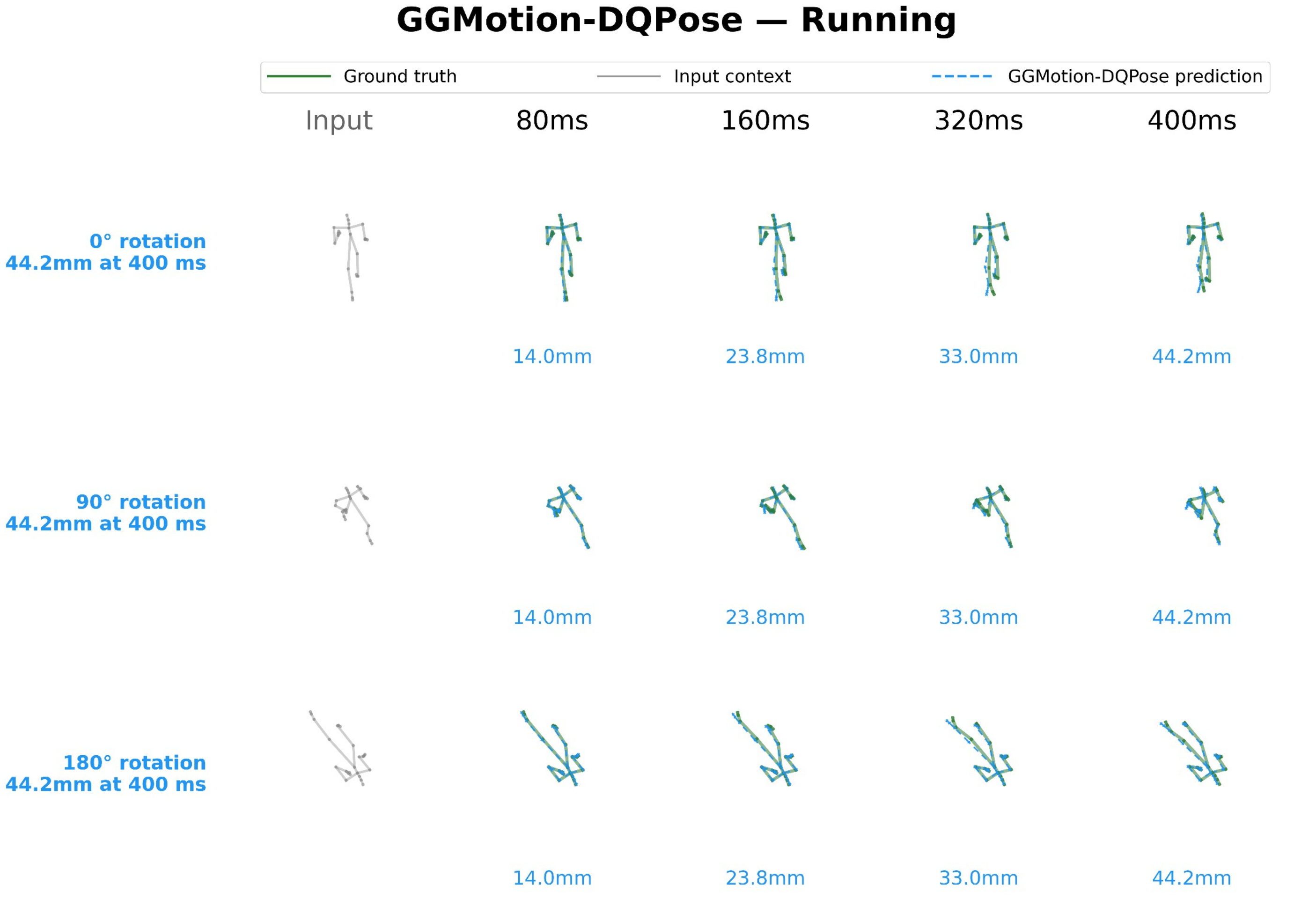
6.13b Rotation Walking: XYZ



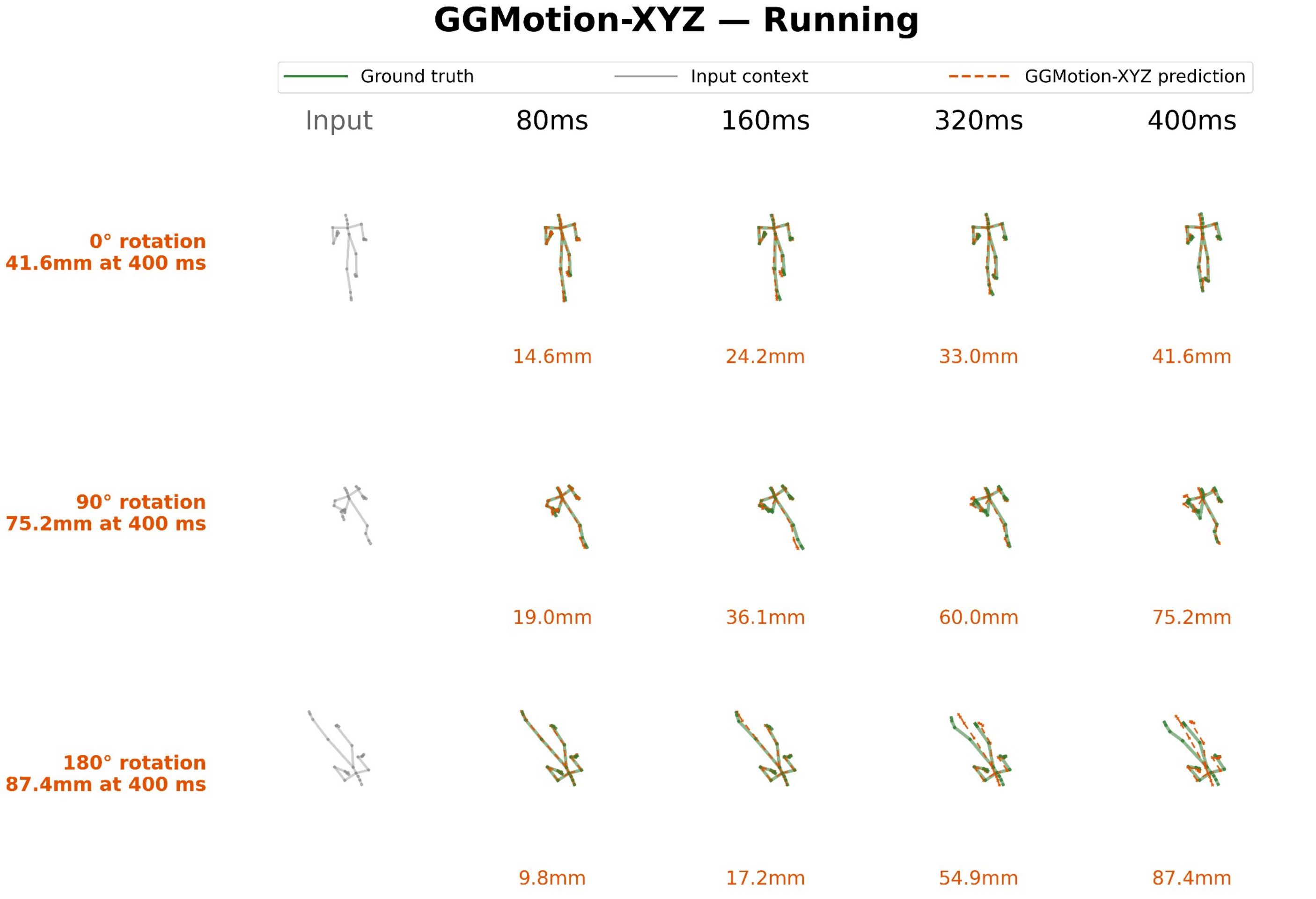
Prediction geometry changes with the angle. XYZ degradation 24.3mm to 52 mm

Walking is strongly direction-dependent. Direction is encoded in the Cartesian stream.

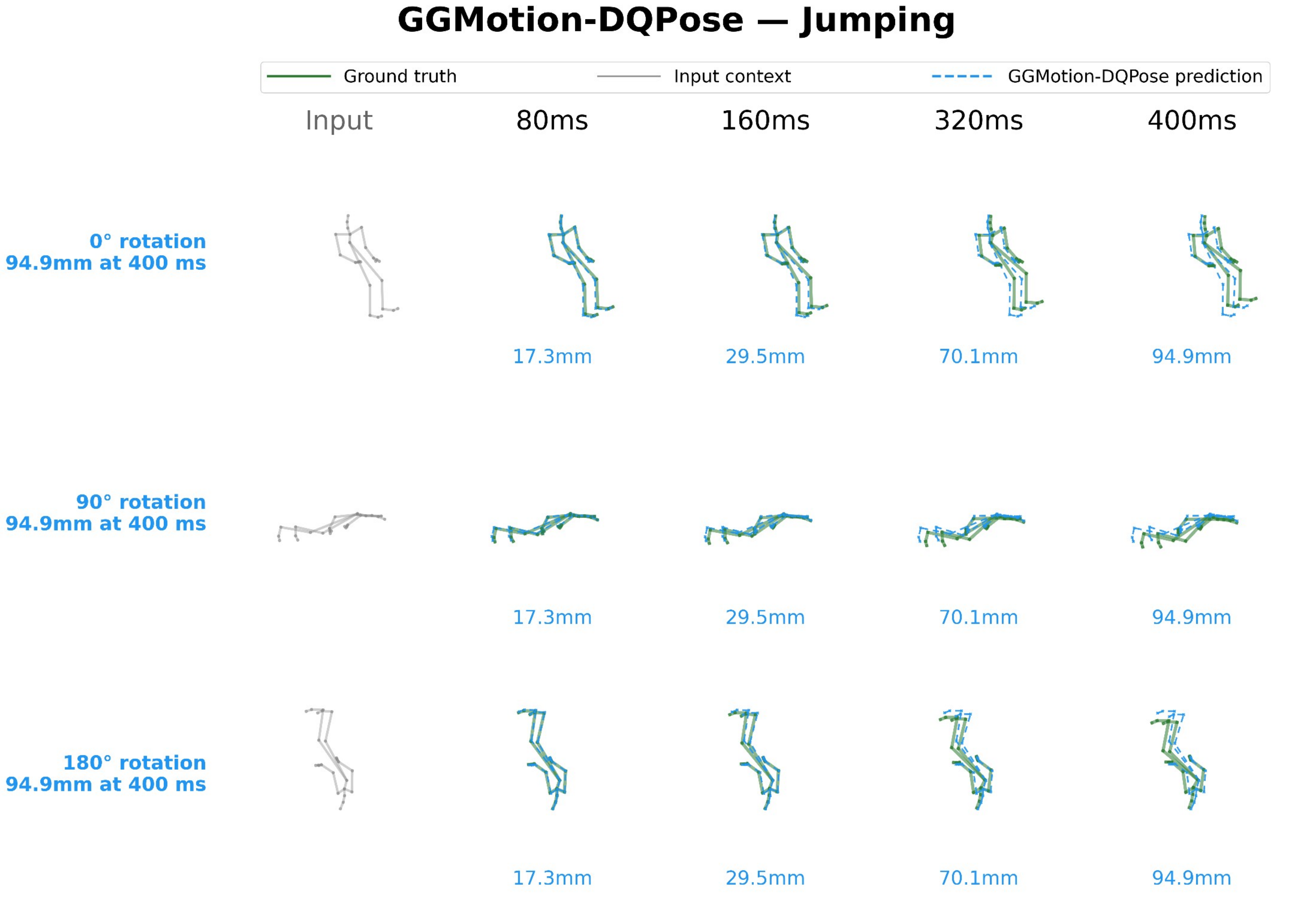
6.14a Rotation Running: DQPose



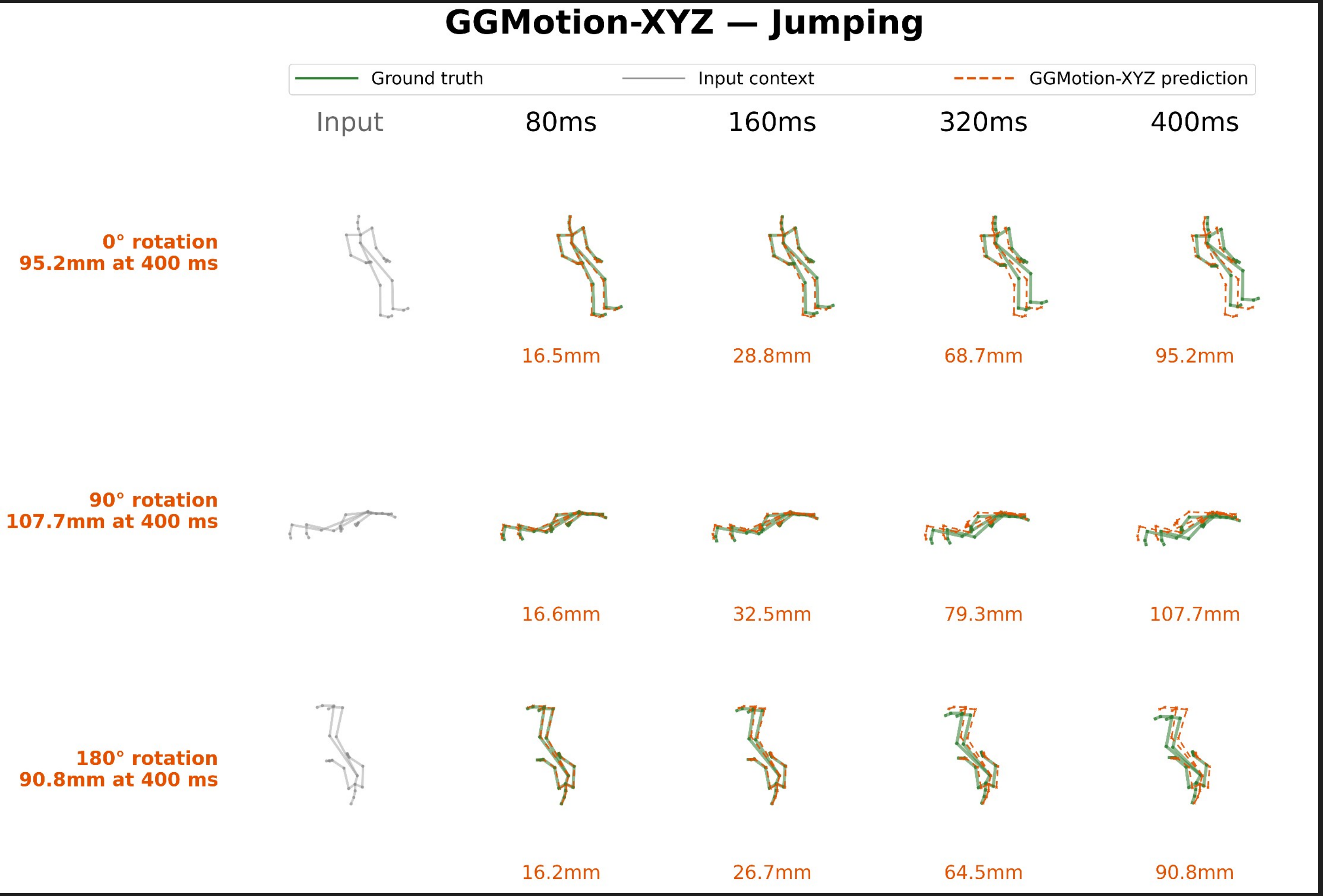
6.14b Rotation Running: XYZ



6.15a Rotation Jumping: DQPose



6.15b Rotation Jumping: XYZ



6.16 Equivariance Overlay: Δ and Divergence

Diagnostic. For each sequence: apply rotations of 0°, 90°, 180° to the input. Predict in the rotated frame. Map predictions back to the original frame by applying the inverse rotation.

For an equivariant model the three mapped-back predictions **coincide exactly**.

Δ MPJPE. Change in MPJPE under the rotation, measured against ground truth. DQPose: Δ = 0. XYZ: Δ > 0 when the prediction degrades.

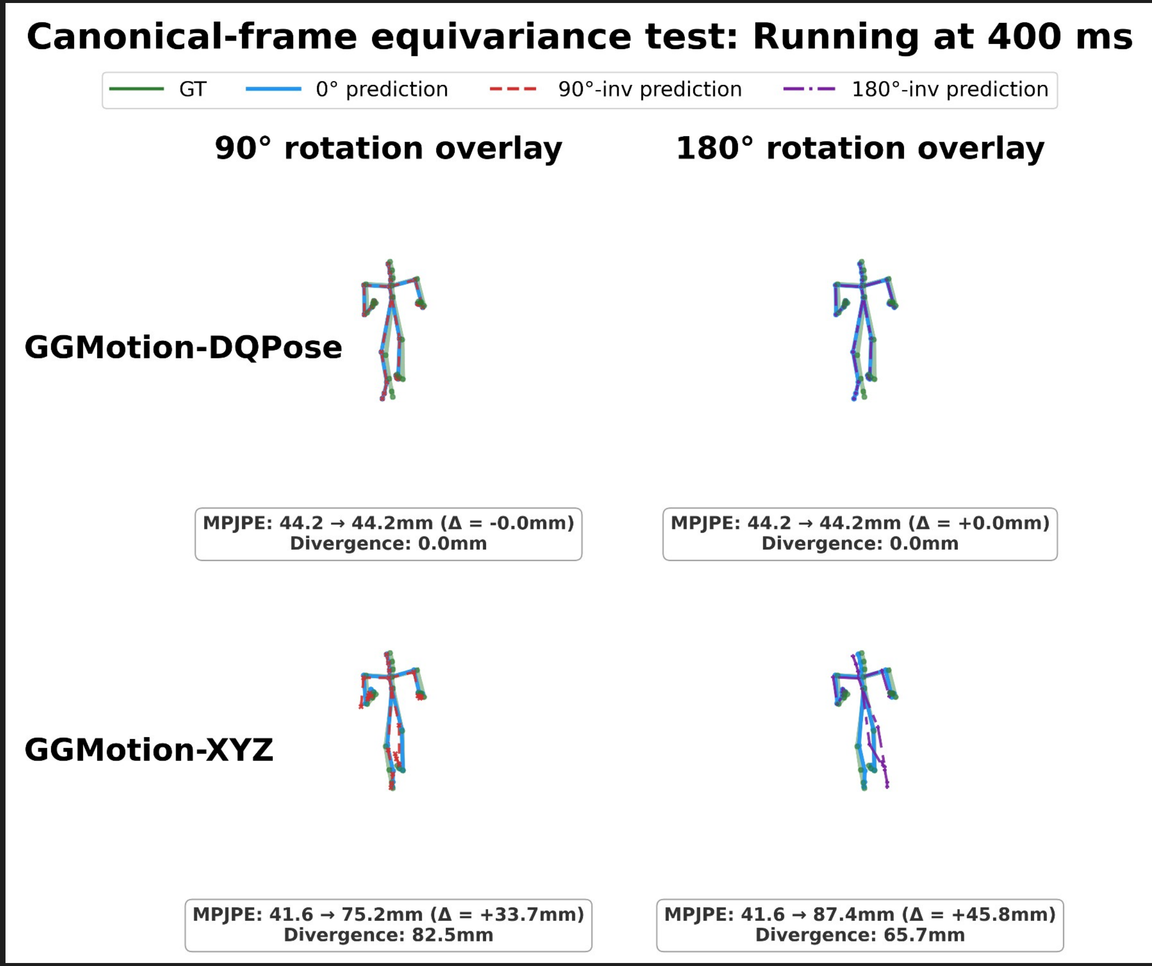
Divergence. Mean per-joint distance between the 0° prediction and the inverse-mapped rotated prediction.

Low divergence: the model produced the same physical motion in a different frame.

High divergence: the prediction itself changed, not only its coordinates.

Why both? MPJPE alone can hide which way the prediction changed. Divergence directly tests prediction geometry.

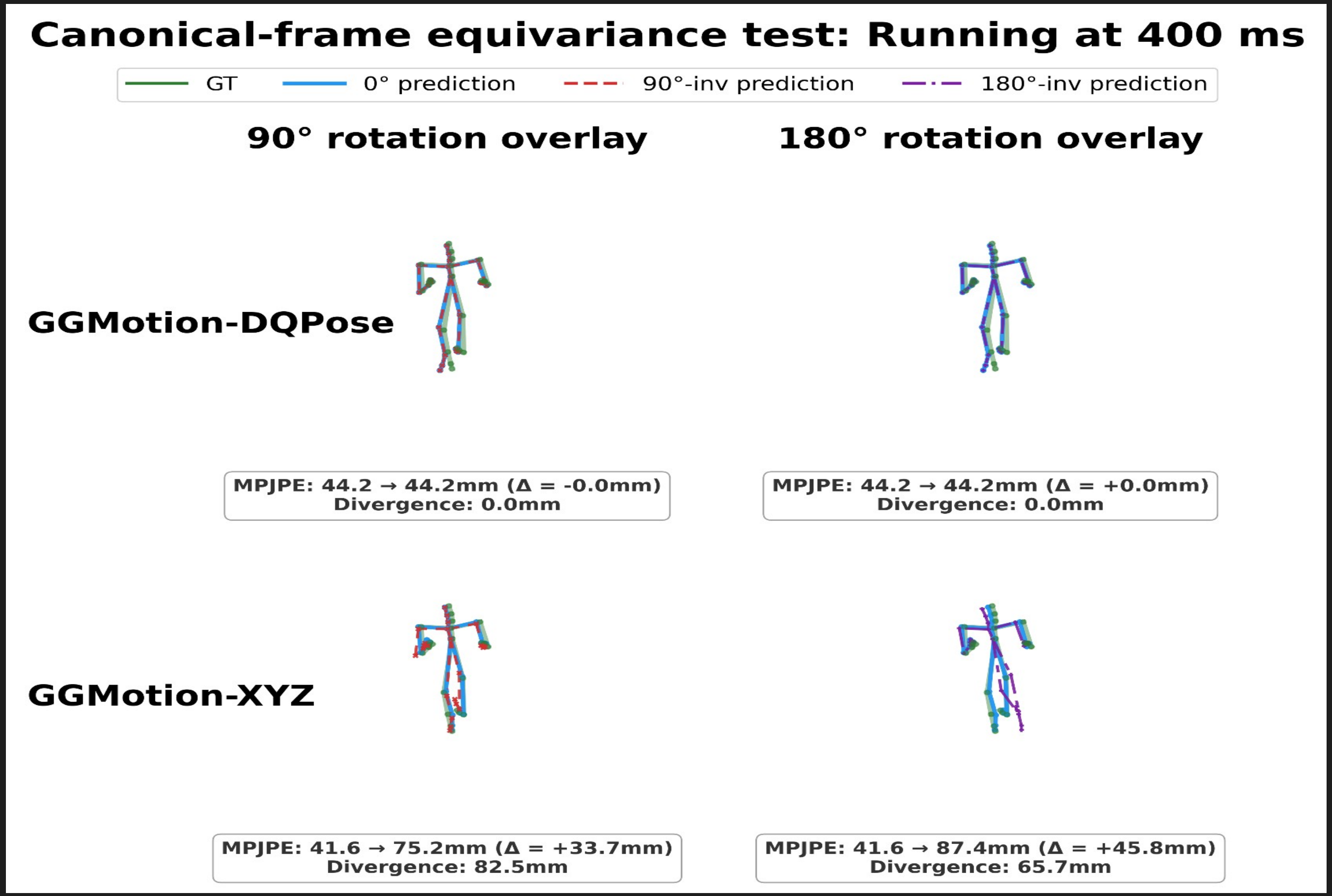
6.17 Equivariance: Running (Figure)



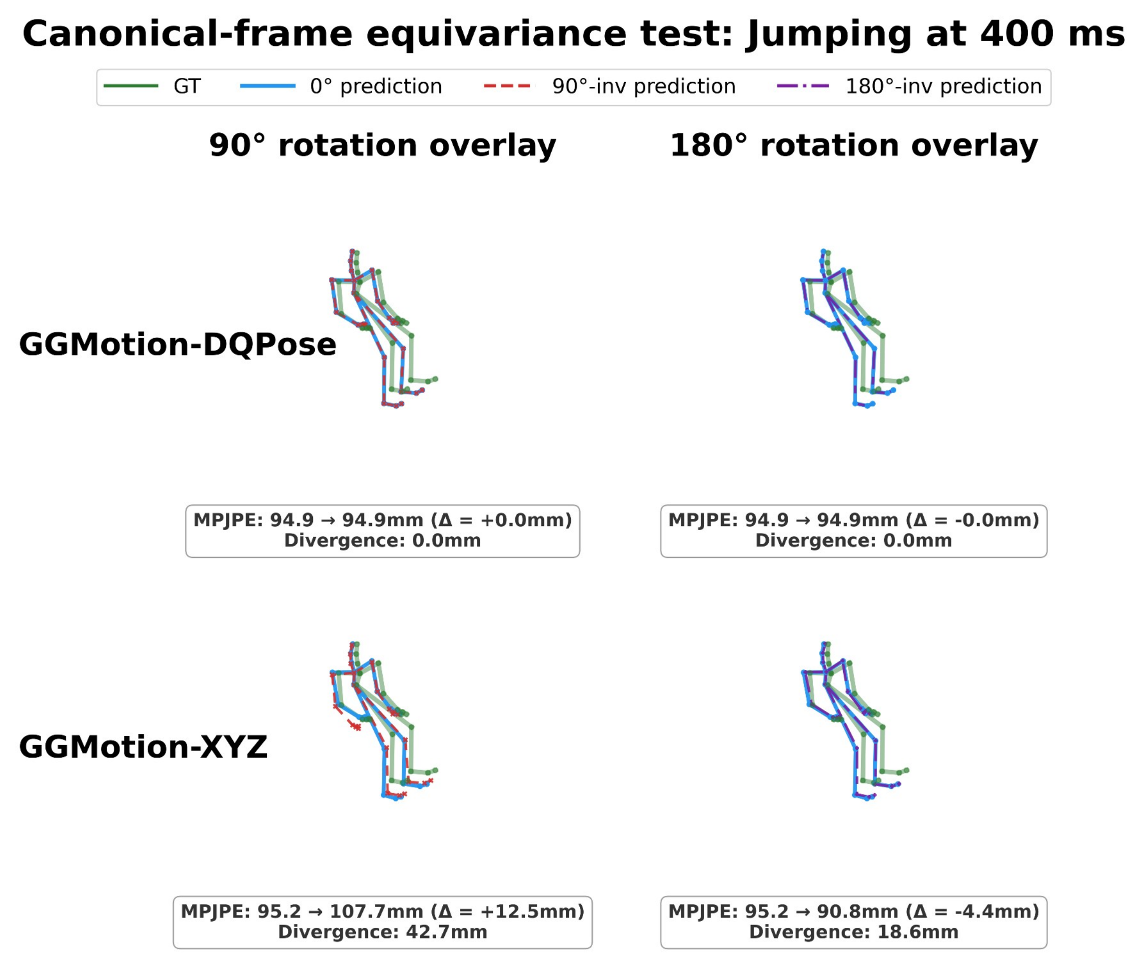
DQPose (top): three mapped-back predictions overlap to numerical precision. Divergence near 0.

XYZ (bottom): visible divergence. Mapped-back predictions are geometrically different poses.

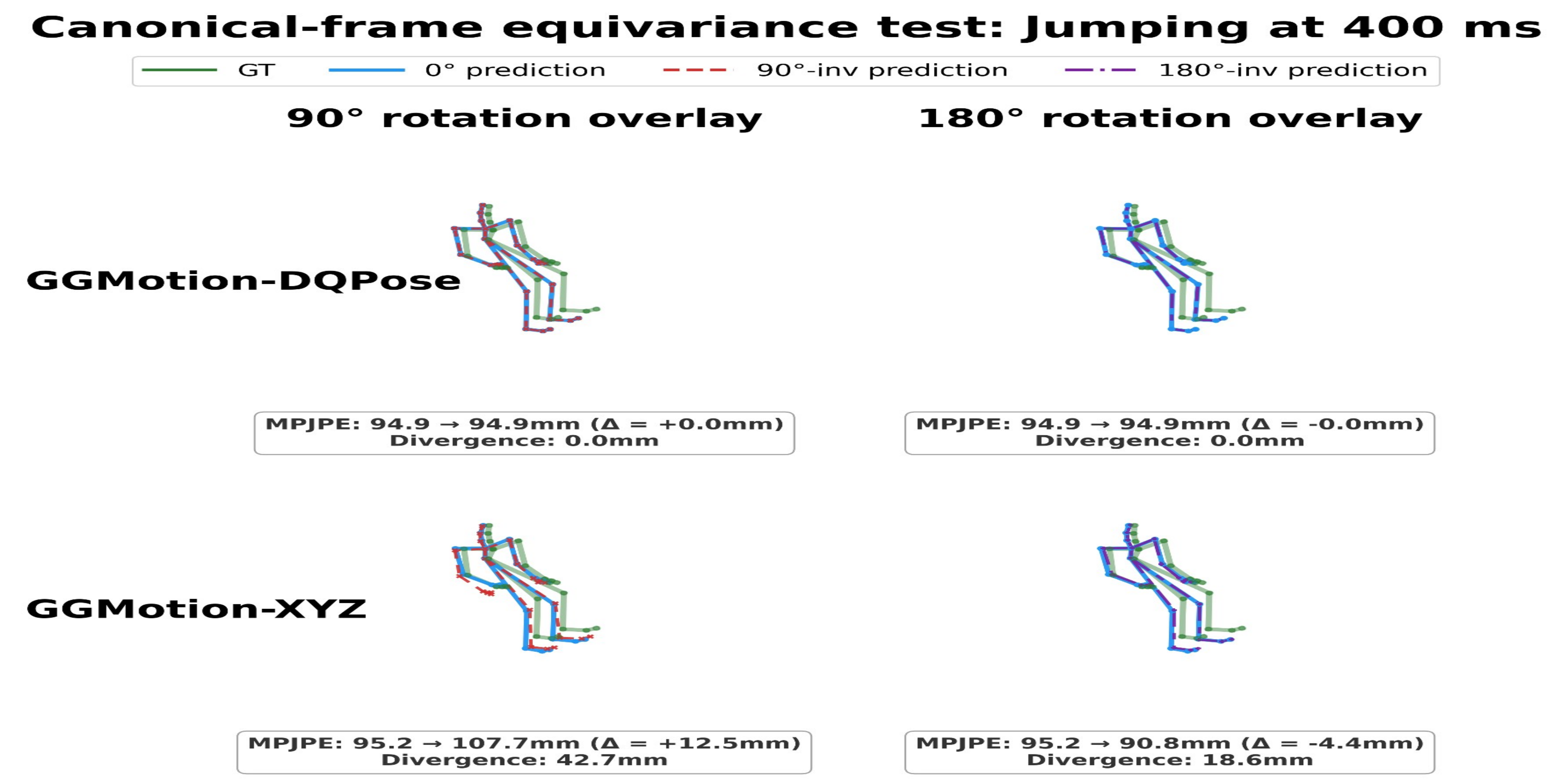
6.18 Equivariance: Running (Video)



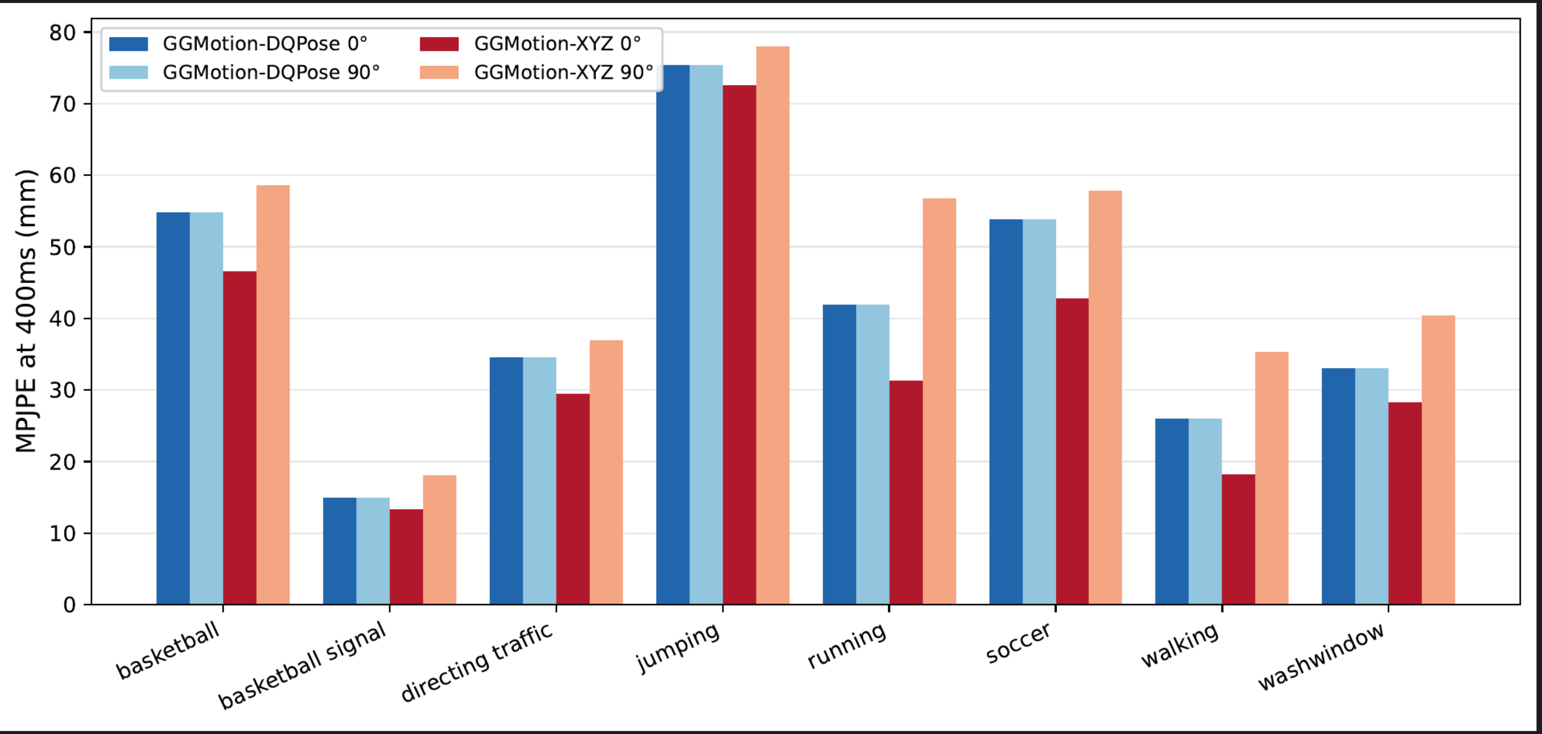
6.19 Equivariance: Jumping (Figure)



6.20 Equivariance: Jumping (Video)



6.21 Per-Action Robustness Plot



DQPose: flat bar on every action. **0% change**.

XYZ: action-dependent.

Largest: walking **+94%**, running **+81%**.

Smallest: jumping **+8%**.

Average: **+35%**.

6.22 Per-Action Robustness Table

Action	DQ 0°	DQ 90°	XYZ 0°	XYZ 90°	Δ XYZ
basketball	54.9	54.9	46.6	58.7	+26%
basketball signal	15.0	15.0	13.3	18.1	+36%
directing traffic	34.6	34.6	29.5	37.0	+25%
jumping	75.4	75.4	72.5	78.0	**+8%**
running	41.9	41.9	31.3	56.7	**+81%**
soccer	53.8	53.8	42.8	57.9	+35%
walking	26.1	26.1	18.2	35.3	**+94%**
washwindow	33.1	33.1	28.3	40.4	+43%
Average	**41.8**	**41.8**	**35.3**	**47.8**	**+35%**

7 Discussion

7.1 Interpretation of the Tradeoff

1.) Accuracy: 400 ms controlled comparison. GGMotion-XYZ at 35.3 mm. GGMotion-DQPose at 41.8 mm.

The accuracy gap could mean DQPose is harder to optimize, or that pose-DQ is the wrong inductive bias for a position metric.

XYZ predicts directly the quantity scored by the dominant loss. DQPose must learn through FK pose DQs, log coordinates, exp decoding, reference composition.

2.) robustness result is the reason DQPose is scientifically useful. Under SE(3) frame change, DQPose stays constant. XYZ degrades.

This is not **DQ beats XYZ** in the benchmark sense.

It is a **measured representation tradeoff**.

7.2 SE(3) by Construction vs Alternatives

Data augmentation. Random rotations and translations during training. Function class still free to fit the symmetry approximately. Continuous-group coverage never complete. Spends capacity on a property that can be enforced algebraically.

Centroid subtraction. Removes some absolute translation dependence. Does not guarantee SE(3) equivariance through the full layer stack. GGMotion-XYZ uses centroid subtraction yet still degrades under translation.

Learned invariance. Published CMU baselines evaluated only in the dataset frame. May learn accidental robustness, but it is unmeasured unless perturbation tests are run.

Architectural equivariance is the only option that gives a guarantee **outside the sampled training frames**.

7.3 Limitations of the Evidence

Accuracy gap of 6.5 mm. Largest limitation. DQPose trails XYZ on canonical MPJPE under this backbone and loss. Contribution is robustness and representation analysis.

Single benchmark. CMU 8-action only. Human3.6M not evaluated. The equivariance proof is architectural; the accuracy tradeoff may change with dataset size, joint topology, action diversity.

Backbone-preserving design. Staying close to GGMotion keeps the comparison clean. May underuse the DQ representation.

Position-based benchmark pressure. MPJPE is the dominant pressure. Necessary for comparability. May not be the loss that best exploits pose dual quaternions.

Short-term deterministic setting. Predicts a fixed 10-frame future. SE(3) property would still be desirable for longer or stochastic settings but is not measured there.

7.4 Broader Implications

A geometrically correct representation can change what a model **guarantees**, even when it does not improve the canonical benchmark number.

Coordinate frame arbitrary or changing between sensors. DQPose property matters. Same motion in a different frame produces the correspondingly transformed prediction.

Evaluation frame fixed, canonical MPJPE the only objective. Cartesian baseline is the better choice.

Useful beyond motion prediction. Many real systems, robots, tools, and rigid mechanical parts, are naturally described by position + orientation, not just XYZ points. Building the symmetry into the representation removes a class of errors that data alone may not eliminate.

8 Conclusion

8.1 Summary of Contributions

1. **Faithful GGMotion-XYZ reproduction.** 35.3 mm at 400 ms, within 0.6 mm of the published 34.7 mm. Strong controlled baseline.

2. **GGMotion-DQPose: a DQ-native counterpart.** FK-derived pose DQs, reference-pose-relative se(3) log coordinates, original GGMotion spatio-temporal pattern, DQ decoding before position loss. 41.8 mm at 400 ms.

3. **Robustness measurement.** Under held-out test-time rotations up to 180° and translations up to 5 m, DQPose stays constant. XYZ rises to 47.8 mm under rotation and 2074 mm under 5 m translation.

Contribution. A measured accuracy-robustness tradeoff with a clear algebraic mechanism.

8.2 Future Work

Representation-aware loss design. Balance decoded positions, pose geodesics, angular and translational velocity, bone structure without letting any term dominate. Possible path to closing the 6.5 mm gap.

Broader benchmarks. Human3.6M is the immediate next step. SE(3) proof transfers. Tradeoff must be measured.

Stochastic and long-horizon prediction. Extend GGMotion-DQPose to longer futures and stochastic models (e.g. diffusion), with noise defined on the DQ manifold via exp/log rather than unconstrained XYZ.

8.3 A Measured Representation Tradeoff

GGMotion-XYZ

Canonical 400 ms: **35.3 mm**

Under 90° rotation: **47.8 mm** (+35%)

Under 5 m translation: **2074 mm**

Frame-dependent. Function tied to the dataset coordinate frame.

GGMotion-DQPose

Canonical 400 ms: **41.8 mm** (6.5 mm behind)

Under 90° rotation: **41.8 mm** (0%)

Under 5 m translation: **41.8 mm** (0%)

Frame-independent. SE(3) frame change cancels algebraically before the backbone.

Contribution. A DQ-native counterpart with an empirical accuracy-robustness tradeoff and a clear algebraic mechanism.